ELLIPTIC CURVE COUNTS AND GENERALIZED WEINSTEIN CONJECTURES

Abstract. We use elliptic curve Gromov–Witten theory to probe various generalizations of the Weinstein conjecture. For instance, we prove new cases of Mazzucchelli's generalization of the Weinstein conjecture, and versions of Sandon's translated points conjecture. We also state an analogue of the Weinstein conjecture in lcs geometry, which implies it, and discuss various partial verifications. A counterexample for a stronger form of this conjecture is given. Along the way we give an intriguing formula for certain $\mathbb{Q}$-valued elliptic Gromov–Witten curve counts in terms of the dynamical Fuller index, which might be of independent interest. The paper aims to be self-contained with no knowledge beyond basic Gromov–Witten theory required.

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1. Introduction

The Weinstein conjecture asserts that the Reeb vector field of any contact form on a closed manifold has a closed orbit [43]. It has been very influential in contact geometry. Attacks on it have produced a range of techniques, most notably the embedded contact homology of Hutchings [18]. In dimension 3 it was proved in full by Taubes [41], using a connection to Seiberg-Witten theory. We study some natural generalizations of the Weinstein conjecture. We argue that one natural home for such a generalization is locally conformally symplectic (lcs) geometry, and more specifically what we call Liouville-lcs geometry, Section 1.1. Our methods are based on counts of elliptic curves in lcs manifolds. They are local in nature: they apply to structures sufficiently close to certain model structures. However, there is strong evidence that a global theory exists. The main evidence is [39], which proves topological invariance of certain rational closed geodesic counts on surfaces. These geodesic counts directly correspond to Gromov-Witten counts of certain rational elliptic curves, of the type studied here, in unit cotangent bundle of oriented surfaces. The topological invariance then translates to invariance of the latter counts under Liouville-lcs equivalences, Definition 1.11, cf. Theorem 6.14. We leave exploration of this to future study.

We will start with the more basic story of fixed Reeb strings of strict contactomorphisms. This will turn out to be closely tied to counts of elliptic curves in certain associated lcs manifolds. From fixed Reeb strings, we are then led to cases of Mazzucchelli's generalization of the Weinstein conjecture, Sandon's translated points, and then to the formulation of the "conformal symplectic Weinstein conjecture".

Let (C^{2n+1},λ) be a closed contact manifold, Terminology 2.2, with λ a contact form, that is a one form s.t. $\lambda\wedge(d\lambda)^n\neq 0$. Denote by R^λ the Reeb vector field satisfying:

$$d\lambda(R^\lambda,\cdot)=0,\quad \lambda(R^\lambda)=1.$$

Recall that a ***closed λ-Reeb orbit*** (or just Reeb orbit when λ is implicit) is a smooth map

$$o:(S^1=\mathbb{R}/\mathbb{Z})\to C$$

such that

$$o'(t)=cR^\lambda(o(t)),$$

with $o'(t)$ denoting the time derivative, for some $c>0$ called period. Let $S(R^\lambda,\beta)$ denote the space of all closed Reeb orbits in free homotopy class β, with its compact open topology. Set

$$\mathcal{O}(R^\lambda,\beta):=S(R^\lambda,\beta)/S^1,$$

where S^1 is acting naturally by reparametrization $(\theta\cdot o)(t)=o(t+\theta)$. Its elements are called ***Reeb strings***; they are the orbit strings of Appendix A for $X=R^\lambda$.

Definition 1.1. Let ϕ be a strict contactomorphism of a contact manifold (C,λ), Terminology 2.2. As $\phi^*\lambda=\lambda$, $\phi_*R^\lambda=R^\lambda$, so $o\mapsto\phi\circ o$ takes closed Reeb orbits to closed Reeb orbits of the same period, commutes with reparametrization, and induces a map $\mathcal{O}(R^\lambda,\beta)\to\mathcal{O}(R^\lambda,\phi_*\beta)$. A ***fixed Reeb string of ϕ in class*** β is an $o\in\mathcal{O}(R^\lambda,\beta)$ with $\phi\circ o=o$; then $\phi_*\beta=\beta$.

Here and below, "homotopic to the id via strict contactomorphisms" means that $\phi=\phi_1$ for a smooth family $\{\phi_t\}_{t\in[0,1]}$ of strict contactomorphisms with $\phi_0=\mathrm{id}$.

Theorem 1.2 (Special case of Theorem 1.7, proved in Section 7)**.** *For any contact form λ on S^{2n+1}, $n\geq 1$, sufficiently C^1 nearby to λ_{st} (the standard contact form), any strict contactomorphism ϕ of (S^{2n+1},λ) homotopic to the* id *via strict contactomorphisms has a fixed Reeb string in the trivial class.*

Let ϕ be a strict contactomorphism of (C,λ). Following Mazzucchelli [22], a ***ϕ-invariant Reeb orbit*** is a trajectory γ of R^λ s.t. for some $T\neq 0$ and all $t\in\mathbb{R}$, $\phi(\gamma(t))=\gamma(t+T)$. Hence id-invariant Reeb

orbits are precisely closed Reeb orbits. Note also that given a fixed Reeb string of ϕ, represented by γ, the latter is also clearly a ϕ-invariant Reeb orbit.

Conjecture 1.3 (Strict form of Mazzucchelli [22, Conjecture 1.2])**.** *For any (C,λ) with C closed, every strict contactomorphism ϕ homotopic to the identity via strict contactomorphisms admits a ϕ-invariant Reeb orbit.*

Mazzucchelli [22, Conjecture 1.2] only requires ϕ to preserve λ and to be contact isotopic to the identity, not necessarily through strict contactomorphisms; the above is the special case that we can address. Theorem 1.2, and more generally Theorem 1.8 below, give:

Corollary 1.4. *Conjecture 1.3, the strict form of Mazzucchelli's generalization of the Weinstein conjecture* [22, Conjecture 1.2]*, holds for C^1 perturbations of (S^{2n+1},λ_{st}), $n\geq 1$. Furthermore, it holds for every (C,λ) satisfying the hypothesis of Theorem 1.8 below, in particular for the contact forms of Theorem 1.7.*

There is a partial connection of Theorem 1.2 with the theory of translated points.

Definition 1.5 (Sandon [36])**.** Given a general contactomorphism ϕ of (C,λ), a point $p\in C$ is called a ***translated point*** provided that $\phi^*\lambda(p)=\lambda(p)$ and $\phi(p)$ lies on the λ-Reeb flow line passing through p.

A fixed Reeb string for ϕ in particular determines a special translated point of ϕ (one for each point on the image of the fixed Reeb string). Hence we get:

Corollary 1.6. *Let ϕ be a strict contactomorphism of (S^{2n+1},λ), $n\geq 1$, homotopic to the* id *via strict contactomorphisms, for λ sufficiently C^1 nearby to λ_{st}. Then ϕ has a non-empty continuum of translated points.*

Note that the general form of Sandon's conjecture on existence of translated points [36] has counterexamples on S^{2n+1} for the standard contact form λ_{st}, see Cant [6]. However, it likely holds for any strict contactomorphism of (S^{2n+1},λ) representing the standard contact structure.

Theorem 1.7 (Proof in Section 7)**.** *Suppose that (C,λ) is closed and Morse–Bott, Appendix A, and some connected component $N\subset\mathcal{O}(R^\lambda,\beta)$, with the union of the images of its orbits orientable, has non-vanishing orbifold Euler characteristic $\chi^{orb}(N)$, defined in Appendix A. Then for any contact form λ' on C, sufficiently C^1 nearby to λ, any strict contactomorphism ϕ of (C,λ'), homotopic to the* id *via strict contactomorphisms, has a class β fixed Reeb string.*

Both of the theorems above are consequences of the following theorem. For more details on the Fuller index see Appendix A. Let λ be a contact form on a closed manifold C, $N\subset\mathcal{O}(R^\lambda,\beta)$ open and compact, and let $i(N,R^\lambda,\beta)\in\mathbb{Q}$ denote the Fuller index. For example, if λ is Morse–Bott (see [4]) and N is a connected component of $\mathcal{O}(R^\lambda,\beta)$, such that the union of the images of the orbits in N is an orientable submanifold of C, then N is naturally a closed orbifold and

$$i(N,R^\lambda,\beta)=\pm\chi^{orb}(N),$$

for $\chi^{orb}(N)$ the orbifold Euler characteristic of N, see (A.3) in Appendix A, cf. [38, Section 2.1.1, proof of Lemma 2.13]. If all the orbits in N have the same multiplicity m, Appendix A, then N is a closed manifold and $\chi^{orb}(N)=\chi(N)/m$, for $\chi(N)$ the Euler characteristic. Examples with $\chi^{orb}(N)\neq 0$ are the standard contact form λ_{st} on S^{2n+1} and the unit cotangent bundle of a round sphere; see Bourgeois [4] for more.

Theorem 1.8 (Proof in Section 7)**.** *Let (C,λ) be a closed contact manifold, and let $N \subset \mathcal{O}(R^\lambda,\beta)$ be open and compact with $i(N,R^\lambda,\beta)\neq 0$. Then any strict contactomorphism $\phi:(C,\lambda)\to(C,\lambda)$, homotopic to the* id *via strict contactomorphisms, has a class β fixed Reeb string $o\in N$.*

The index assumption of the theorem thus holds for the Morse–Bott examples above. In this Morse–Bott case, if the underlying topological space of N has $\chi(N)\neq 0$ (equivalent to $\chi^{orb}(N)\neq 0$ when all the orbits in N have the same multiplicity), the theorem may be verified by elementary considerations. Indeed, ϕ as above induces a homeomorphism ϕ_N of N, homotopic to the identity, hence with non-zero Lefschetz number, so that in this case the result follows by the Lefschetz fixed point theorem.

In general a compact open component $N\subset\mathcal{O}(R^\lambda,\beta)$ may not be a finite simplicial complex, or indeed any kind of topological space to which the classical Lefschetz fixed point theorem may apply. Also the relationship of $i(N,R^\lambda,\beta)$ with $\chi(N)$ breaks down in general as $i(N,R^\lambda,\beta)$ is partly sensitive to the dynamics of R^λ.

1.1. **Liouville-lcs geometry.** For preliminaries on lcs geometry see Section 2; in particular lcs pairs (λ,α) and their lcs forms $\omega=d_\alpha\lambda$ are defined in Definition 2.5.

Definition 1.9. Let (λ,α) be an lcs pair on M and $\omega=d_\alpha\lambda$, for d_α the Lichnerowicz differential (2.3). Define X_λ by $\omega(X_\lambda,\cdot)=\lambda$ and X_α by $\omega(X_\alpha,\cdot)=\alpha$, and generally X_η by $\omega(X_\eta,\cdot)=\eta$ for a 1-form η. Let $\mathcal{D}$ denote the (generalized) distribution spanned by X_α, X_λ, meaning $\mathcal{D}(p):=\operatorname{span}(X_\alpha(p),X_\lambda(p))$. This will be called the ***canonical distribution***.

Definition 1.10. An lcs pair (λ,α) on M satisfies the ***Reeb condition*** if

$$\lambda(X_\alpha)>0.$$

An lcs pair satisfying the Reeb condition is called a ***Reeb pair***, and a manifold with a Reeb pair a ***Liouville-lcs manifold***.

Definition 1.11. Two Reeb pairs (λ_0,α_0) on M_0 and (λ_1,α_1) on M_1 are ***Liouville-lcs equivalent*** if there is a diffeomorphism $\phi:M_0\to M_1$ such that $\phi^*(\lambda_1,\alpha_1)$ and (λ_0,α_0) are joined by a smooth family of Reeb pairs on M_0; such ϕ is called a ***Liouville-lcs equivalence***.

Liouville-lcs equivalences compose and are invertible, by pulling back, reparametrizing and reversing the families. So Reeb pairs with Liouville-lcs equivalences form a groupoid, the ***Liouville-lcs category***; Liouville isomorphisms, Definition 2.7, are the morphisms for which the family may be taken constant.

Conceptually this paper is about Liouville-lcs geometry, not lcs geometry in generality. An lcs pair (λ,α) is the lcs analogue of a Liouville structure, with Liouville field X_λ, $\iota_{X_\lambda}\omega=\lambda$. The lcs-fication of (C,λ), Example 2.4, is the symplectization $(\mathbb{R}\times C, d(e^r\lambda))$ divided by $r\mapsto r+1$, with $\theta=-r$. Its lcs form is the invariant form $e^{-r}d(e^r\lambda)=d\lambda+dr\wedge\lambda$. The Liouville field ∂_r descends to $X_\lambda=-\partial_\theta$. The almost complex structure J^λ of Example 4.6 is the standard cylindrical one, $J^\lambda\partial_r=R^\lambda$. The leaves L of the foliation $\ker\alpha$, which we call the ***Lee foliation*** (a foliation where $\alpha\neq 0$; under the Reeb condition α vanishes nowhere, as $X_\alpha\neq 0$), satisfy $\omega|_L=d(\lambda|_L)$, and their characteristic line field is spanned by X_α, as $X_\alpha\in\ker\alpha$ is ω-orthogonal to $\ker\alpha$. As $\alpha(X_\lambda)=-\lambda(X_\alpha)$, the Reeb condition implies that the Liouville field is transverse to the Lee foliation, that is $\lambda(X_\alpha)$ vanishes nowhere. On a closed connected M, oriented by ω^n, the two are equivalent: $d(\lambda\wedge\omega^{n-1})=(1-\lambda(X_\alpha))\omega^n$, so $\int_M\lambda(X_\alpha)\omega^n=\int_M\omega^n>0$ by Stokes. Where $\alpha\neq 0$, transversality is equivalent to $\lambda|_L$ being a contact form on each leaf L, with Reeb field $X_\alpha/\lambda(X_\alpha)$, by the computation in the proof of Proposition 1.18.

1.2. **Conformal symplectic Weinstein conjecture.** The goal is now to describe one analogue of the Weinstein conjecture in Liouville-lcs geometry.

1.2.1. *Reeb 2-curves.* The canonical distribution $\mathcal{D}$ of an lcs pair is one analogue of the Reeb distribution on contact manifolds. A Reeb 2-curve, as defined ahead, will be a certain kind of singular surface tangent to $\mathcal{D}$, and so is a kind of 2-dimensional analogue of a Reeb orbit.

Definition 1.12. Let Σ be a closed nodal Riemann surface (the set of nodes can be empty). Let $u : \Sigma \to M$ be a smooth map and let $\widetilde{u} : \widetilde{\Sigma} \to M$ be its normalization (see Definition 4.2). We say that u is a ***Reeb 2-curve*** of an lcs pair (λ, α) on M, if the following is satisfied:

(1) For each $z \in \widetilde{\Sigma}$ at which $d\widetilde{u}(z) : T_z\widetilde{\Sigma} \to T_{\widetilde{u}(z)}M$ is non-zero, $\dim \mathcal{D}(\widetilde{u}(z)) = 2$ and $d\widetilde{u}(z)(T_z\widetilde{\Sigma}) = \mathcal{D}(\widetilde{u}(z))$.

(2) $0 \neq u^*[\alpha] \in H^1(\Sigma, \mathbb{R})$.

Example 1.13. The simplest example of a Reeb 2-curve in a manifold M with an lcs pair (λ, α), in the case $\mathcal{D}$ is a true 2-dimensional distribution (for example if (λ, α) is first kind, Definition 2.6, or a Reeb pair, Definition 1.10), is a closed oriented immersed surface $u : \Sigma \to M$ tangent to $\mathcal{D}$ with $u^*[\alpha] \neq 0$. The last condition is automatic when $\lambda(X_\alpha) \neq 0$ along u, in particular in the first kind case: then ω is non-degenerate on $\mathcal{D}$, so $(u^*\lambda, u^*\alpha)$ is an lcs pair on the closed surface Σ, and Lemma 3.6 applies. However, it will be necessary to consider more generalized curves.

It is tempting to conjecture that every closed manifold with an lcs pair has a Reeb 2-curve, in analogy to the Weinstein conjecture. However this is false:

Theorem 1.14 (Proof in Section 7)**.** *Let (T^2, g_{st}) be the 2-torus with its standard flat metric. Let $M_{\widetilde{\phi},1}$ be the mapping torus, Example 2.9, of the unit cotangent bundle S^*T^2 with its canonical contact form and of the strict contactomorphism $\widetilde{\phi}$ of it induced by an isometry $\phi : (T^2, g_{st}) \to (T^2, g_{st})$, no power of which fixes the image of any closed geodesic (for instance a translation by (a, b) with $1, a, b$ linearly independent over $\mathbb{Q}$). Then $M_{\widetilde{\phi},1}$ has no Reeb 2-curves, although $(\lambda_{\widetilde{\phi}}, \alpha)$ is a Reeb pair.*

1.2.2. *Reeb 1-curves.* We can repair the tempting conjecture above by working with a more flexible notion, which is as follows.

Definition 1.15. A smooth map $o : S^1 \to M$ is a ***Reeb 1-curve*** of an lcs pair (λ, α) on M if

$$\forall t \in S^1 : (\lambda(o'(t)) > 0) \wedge (o'(t) \in \mathcal{D}(o(t))).$$

The following is elementary; its proof, in Section 3, uses Theorem 3.4, proved in Section 7.

Proposition 1.16. *Every closed Liouville-lcs surface has a Reeb 1-curve.*

We will refer to the following as the conformal symplectic Weinstein conjecture.

Conjecture 1.17. *Every closed Liouville-lcs manifold of dimension 4 has a Reeb 1-curve.*

Let (λ, α) be a Reeb pair on a closed connected M. For scale integral α the leaves of $\ker \alpha$ are the components of the fibers of a fibration $M \to S^1$, which are closed contact manifolds, and Conjecture 1.17 follows from the Weinstein conjecture for them, Proposition 1.18 and Corollary 1.19. For scale irrational α no leaf is compact: a foliation without holonomy on a closed manifold, such as $\ker \alpha$, is either a fibration over S^1 or has every leaf dense, Imanishi [19], see also [15, Chapter IX], and in the first case α is scale integral. A Reeb 1-curve replaces a closed characteristic: a loop tangent to $\mathcal{D} = \operatorname{span}(X_\lambda, X_\alpha)$ and positive on λ, which may drift along the Liouville field.

The scale integral case is the following.

Proposition 1.18. *Let (λ,α) be an lcs pair on a closed manifold M^{2n}, with α scale integral: $c\alpha$ has integral periods for some $0\neq c\in\mathbb{R}$, see Definition 2.6. Suppose that $\alpha(X_\lambda)$ vanishes nowhere, that is α vanishes nowhere and X_λ is transverse to $\ker\alpha$. This holds under the Reeb condition, Section 1.1. Then there is a map $p:M\to S^1=\mathbb{R}/\mathbb{Z}$ with $p^*d\theta=c\alpha$, for $d\theta$ as in Example 2.4, and it is a fibration; each fiber $C_t=p^{-1}(t)$ is a closed contact manifold with contact form $\lambda|_{C_t}$, and $X_\alpha|_{C_t}=\lambda(X_\alpha)R_t$ for R_t the Reeb vector field of $\lambda|_{C_t}$. In particular every closed Reeb orbit of $(C_t,\lambda|_{C_t})$ is a Reeb 1-curve of (λ,α).*

Proof. As α vanishes nowhere, p is a submersion, and a fibration as M is closed. On $TC_t=\ker\alpha$ the form $\alpha\wedge\lambda$ vanishes, so $\omega=d\lambda$ there and

$$\lambda\wedge(d\lambda)^{n-1}|_{C_t}=\lambda\wedge\omega^{n-1}|_{C_t}=\frac{1}{n}\iota_{X_\lambda}\omega^n|_{C_t},$$

which vanishes nowhere, as ω^n is a volume form and X_λ is transverse to C_t. So $\lambda|_{C_t}$ is contact. Next

$$\alpha(X_\alpha)=\omega(X_\alpha,X_\alpha)=0,$$

so X_α is tangent to C_t, and

$$\iota_{X_\alpha}d\lambda=\iota_{X_\alpha}\omega+\iota_{X_\alpha}(\alpha\wedge\lambda)=(1-\lambda(X_\alpha))\alpha$$

vanishes on TC_t. Thus $X_\alpha|_{C_t}$ spans $\ker d\lambda|_{C_t}$, and $\lambda(X_\alpha)=-\alpha(X_\lambda)\neq 0$ gives $X_\alpha=\lambda(X_\alpha)R_t$. A closed orbit $o:\mathbb{R}/\mathbb{Z}\to C_t$ of R_t of period $T>0$ has $o'=TR_t\in\mathbb{R}X_\alpha\subset\mathcal{D}$ and $\lambda(o')=T>0$, so it is a Reeb 1-curve. □

Corollary 1.19. *Let (λ,α) be a Reeb pair on a closed 4-manifold M, with α scale integral. Then Conjecture 1.17 holds for (λ,α).*

Proof. By Proposition 1.18 the fibers C_t are closed contact 3-manifolds. Each has a closed Reeb orbit by Taubes [41], and these are Reeb 1-curves. □

Scale integrality holds for every closed 1-form α when $b_1(M)=1$, for instance on $S^3\times S^1$. Thus Conjecture 1.17 is open only for scale irrational α, so only when $b_1(M)\geq 2$. Theorem 1.21 below and Corollary 3.5 are local. They do not use Taubes' theorem, and on $C\times S^1$ with $b_1(C)\geq 1$ they also cover scale irrational α, to which Corollary 1.19 does not apply.

Proposition 1.20. *The analogue of Conjecture 1.17, in all dimensions, implies the Weinstein conjecture: every closed contact manifold (C,λ) has a closed Reeb orbit.*

Proof. Let (λ,α) be the lcs-fication on $M=C\times S^1$ of a closed contact manifold (C,λ), see Example 2.4. Then it satisfies the Reeb condition, as $X_\alpha=(R^\lambda,0)$ and $\lambda(R^\lambda)=1$, see Example 4.6. Suppose that $o:S^1\to M$ is a Reeb 1-curve. By Example 4.6, $\mathcal{D}=\operatorname{span}((R^\lambda,0),(0,\partial_\theta))$, so $o'=a(R^\lambda,0)+b(0,\partial_\theta)$ with $a=\lambda(o')>0$. Hence $(pr_C\circ o)'=aR^\lambda$, and $pr_C\circ o$ is a closed Reeb orbit of (C,λ) after reparametrization to constant speed. □

Theorem 1.21 (Weaker form of Corollary 3.5)**.** *Let λ be a contact form on a closed 3-manifold C with a non-degenerate closed Reeb orbit, that is one whose linearized return map on $\xi=\ker\lambda$ has no eigenvalue 1. Every first kind lcs pair on $C\times S^1$ sufficiently C^∞ close to the lcs-fication (λ,α) has a Reeb 1-curve.*

For $b_1(C)=0$ every closed 1-form on $C\times S^1$ is scale integral, and the theorem, as well as Corollary 3.5, follows from Corollary 1.19. Their content beyond Taubes is the case $b_1(C)\geq 1$, where nearby first kind lcs pairs with scale irrational α' exist, see Remark 2.11.

1.3. **Organization.** Section 2 presents preliminaries on lcs geometry, which should make this paper self-contained and accessible to a general reader. Section 3 states the main results on Reeb curves, Theorems 3.3 and 3.4, and proves Lemma 3.6 and Proposition 1.16. Section 4 introduces charged elliptic curves and the admissible almost complex structures. Section 5 sets up the Gromov–Witten counts of such curves and their invariance under small deformations. Section 6 identifies these counts, for lcs-fications, with the Fuller index. The main theorems are proved in Section 7. Appendix A recalls the Fuller index, and Appendix B the multiplicities of multiply covered curves.

2. Preliminaries on lcs manifolds

Definition 2.1. A locally conformally symplectic manifold, or just an lcs manifold, is a smooth $2n$-fold M with an lcs structure, or lcs form: a non-degenerate 2-form ω, with the property that for every $p \in M$ there is an open $U \ni p$ such that $\omega|_U = f_U \cdot \omega_U$, for some symplectic form ω_U defined on U and some smooth positive function f_U on U.

These kinds of structures were originally considered by Lee in [21], arising naturally as part of an abstract study of "a kind of even dimensional differential geometry", and then further studied by a number of authors, see for instance [2] and [42]. An lcs manifold admits all the interesting classical notions of a symplectic manifold, like Lagrangian submanifolds and Hamiltonian dynamics, while at the same time forming a much more flexible class. For example Eliashberg and Murphy show that if a closed almost complex $2n$-fold M has $H^1(M, \mathbb{R}) \neq 0$ then it admits an lcs form, [8]. Another result of Apostolov–Dloussky [1] is that any compact complex surface with an odd first Betti number admits an lcs form, which tames the complex structure.

To see the connection with the first cohomology group $H^1(M, \mathbb{R})$, mentioned above, let us point out right away the most basic invariant of an lcs form ω, when M has dimension at least 4. We take a cover of M by open sets U_a in which $\omega = e^{f_a} \cdot \omega_a$ for ω_a symplectic. Then we have 1-forms $d(f_a)$ on each U_a, which glue to a well-defined closed 1-form α on M, as shown by Lee, called the ***Lee form***; it is characterized by

$$d\omega = \alpha \wedge \omega.$$

Its class $[\alpha] \in H^1(M, \mathbb{R})$ is the ***Lee class***, and we may denote this 1-form and its cohomology class both by α. On the associated α-covering space $\widetilde{M}$ (the covering of the proof of Lemma 6.1, on which α pulls back to an exact form) the lift $\widetilde{\omega}$ is globally conformally symplectic, and an lcs form on M itself is globally conformally symplectic, that is of the form $e^f \cdot \omega'$ with ω' symplectic, iff its Lee class vanishes. Indeed, if $\alpha = df$ then $d(e^{-f}\omega) = 0$. Conversely, if $\omega = e^f \omega'$ then $\alpha = df$ by the characterization above.

Lcs manifolds can also be understood to generalize contact manifolds. This works as follows. First we have a class of explicit examples of lcs manifolds, obtained by starting with a symplectic cobordism (see [8]) of a closed contact manifold C to itself, arranging for the contact forms at the two ends of the cobordism to be proportional and then gluing the boundary components (after rescaling the form by a positive function, equal to 1 near one end and to a constant near the other, that is a conformal rescaling of the form on the cobordism, to match the boundary conditions).

Terminology 2.2. For us a contact manifold is a pair (C, λ) where C is a manifold and λ a contact form on the $(2n+1)$-fold C: $\forall p \in C : \lambda \wedge (d\lambda)^n(p) \neq 0$. This is not a completely common terminology as usually it is the equivalence class of (C, λ) that is called a contact manifold, where $(C, \lambda) \sim (C, \lambda')$ if $\lambda = f\lambda'$ for f a positive function. (Given that the contact structure, in the classical sense, is co-oriented.) A ***contactomorphism*** between (C_1, λ_1), (C_2, λ_2) is a diffeomorphism $\phi : C_1 \to C_2$ s.t. $\phi^*\lambda_2 = f\lambda_1$ for some $f > 0$. It is called ***strict*** if $\phi^*\lambda_2 = \lambda_1$.

A concrete basic example, which can be understood as a special case of the above cobordism construction, is the following. The operator

$$d_\alpha : \Omega^k(M) \to \Omega^{k+1}(M) \tag{2.3}$$

$d_\alpha \eta := d\eta - \alpha \wedge \eta$ is called the Lichnerowicz differential with respect to the closed 1-form α, and it satisfies $d_\alpha \circ d_\alpha = 0$ so that we have an associated Lichnerowicz chain complex.

Example 2.4 (Banyaga). Let (C, λ) be a contact manifold, $S^1 = \mathbb{R}/\mathbb{Z}$, $d\theta$ the standard non-degenerate 1-form on S^1 satisfying $\int_{S^1} d\theta = 1$. And take $M = C \times S^1$ with the 2-form

$$\omega_\lambda = d_\alpha \lambda = d\lambda - \alpha \wedge \lambda,$$

for d_α the Lichnerowicz differential (2.3), for $\alpha := pr_{S^1}^* d\theta$, $pr_{S^1} : C \times S^1 \to S^1$ the projection, and λ likewise the pull-back of λ by the projection $C \times S^1 \to C$. We call the lcs pair (λ, α) on M as above, Definition 2.5 below, and also (M, ω_λ), the ***lcs-fication*** of (C, λ). This is also a basic example of a first kind lcs pair, Definition 2.6 ahead, as $\ker d\lambda = \operatorname{span}(R^\lambda, \partial_\theta)$.

Definition 2.5. An ***lcs pair*** on M is a pair of 1-forms (λ, α), with α closed, such that $\omega = d_\alpha \lambda$ is non-degenerate; ω is then an lcs form, as locally $\alpha = dg$ and $\omega = e^g d(e^{-g}\lambda)$, cf. the proof of Lemma 3.6; it is called the lcs form of the pair. An lcs form is called ***exact*** if it is the lcs form of some lcs pair.

An lcs pair determines a generalized distribution $\mathcal{V}_\lambda$ on M:

$$\mathcal{V}_\lambda(p) = \{v \in T_p M \,|\, d\lambda(v, \cdot) = 0\},$$

which we call the ***vanishing distribution***. We also define a generalized distribution ξ_λ that is the ω-orthogonal complement to $\mathcal{V}_\lambda$, the ***co-vanishing distribution***. For each $p \in M$, $\mathcal{V}_\lambda(p)$ has dimension at most 2 since $d\lambda - \alpha \wedge \lambda$ is non-degenerate, see the computation following Definition 2.6. If M^{2n} is closed $\mathcal{V}_\lambda$ cannot identically vanish, since $(d\lambda)^n$ cannot be nowhere vanishing by Stokes theorem.

Definition 2.6. Let α be a closed 1-form on M. We call α ***integral*** if its periods are integral, ***scale integral*** if $c\alpha$ is integral for some $0 \neq c \in \mathbb{R}$, and ***scale irrational*** otherwise; these are properties of $[\alpha] \in H^1(M, \mathbb{R})$. An lcs pair (λ, α) is called integral, scale integral or scale irrational if α is. If $\mathcal{V}_\lambda$ is non-zero at each point of M, so that, $d\lambda$ having even rank, it is a smooth 2-distribution, then the lcs pair (λ, α) is called ***first kind***. An exact lcs form ω is called first kind if $\omega = d_\alpha \lambda$ for some first kind lcs pair (λ, α). When $\dim M \geq 4$, α is the Lee form of $d_\alpha \lambda$, as $d(d_\alpha \lambda) = \alpha \wedge d_\alpha \lambda$; for $\dim M \geq 4$ we call any lcs form integral, scale integral or scale irrational if its Lee form is.

For any lcs pair $\mathcal{V}_\lambda \subset \mathcal{D}$, the canonical distribution of Definition 1.9: if $d\lambda(v, \cdot) = 0$ then $\iota_v \omega = \lambda(v)\alpha - \alpha(v)\lambda$, so that $v = \lambda(v) X_\alpha - \alpha(v) X_\lambda$. If the lcs pair is first kind then $\mathcal{V}_\lambda$ is 2-dimensional and $\dim \mathcal{D} \leq 2$, so $\mathcal{V}_\lambda = \mathcal{D}$. Then $X_\alpha \in \ker d\lambda$, while $d\lambda(X_\alpha, \cdot) = (1 - \lambda(X_\alpha))\alpha$, using $\alpha(X_\alpha) = \omega(X_\alpha, X_\alpha) = 0$. Moreover α vanishes nowhere, as $\dim \mathcal{D} = 2$ forces $X_\alpha \neq 0$. Hence $\lambda(X_\alpha) = 1$ everywhere for a first kind lcs pair.

Definition 2.7. A ***conformal symplectomorphism*** of lcs manifolds $\phi : (M_1, \omega_1) \to (M_2, \omega_2)$ is a diffeomorphism ϕ s.t. $\phi^* \omega_2 = e^f \omega_1$, for some f. Note that in this case we have an induced relation (when M_1, M_2 have dimension at least 4):

$$\phi^* \alpha_2 = \alpha_1 + df,$$

where α_i is the Lee form of ω_i, $i = 1, 2$. If $f = 0$ we call ϕ a ***symplectomorphism***. A (conformal) symplectomorphism of lcs pairs (λ_1, α_1), (λ_2, α_2) on M_1 respectively M_2 is a (conformal) symplectomorphism of the corresponding lcs 2-forms. If a diffeomorphism $\phi : M_1 \to M_2$ satisfies $\phi^* \lambda_2 = \lambda_1$ and $\phi^* \alpha_2 = \alpha_1$ we call it a ***Liouville isomorphism*** of the lcs pairs. This is analogous to a strict contactomorphism of contact manifolds.

To summarize, with the above notions we have the following basic points whose proof is left to the reader:

(1) A Liouville isomorphism of lcs pairs (λ_1, α_1), (λ_2, α_2) preserves the first kind condition, and moreover preserves the corresponding vanishing distributions.

(2) A symplectomorphism of lcs forms preserves the first kind condition.

(3) A symplectomorphism, conformal or not, of lcs pairs generally does not preserve the first kind condition, as it only relates the two lcs forms. For instance, for the lcs-fication (λ, α) of (S^3, λ_{st}) and $g : S^3 \to \mathbb{R}$ with $dg(R^\lambda) \neq 0$ somewhere, $(\lambda + dg - g\alpha, \alpha)$ has the same lcs form and is not first kind.

(4) A (conformal) symplectomorphism of first kind lcs pairs generally does not preserve the vanishing distributions. (Similar to a contactomorphism not preserving Reeb distributions.)

(5) A conformal symplectomorphism of lcs forms and of lcs pairs preserves the integrality, scale integrality and scale irrationality conditions, in dimension at least 4, where the Lee form is determined by the lcs form.

Remark 2.8. We say that ω_0 is conformally equivalent to ω_1 if $\omega_1 = e^f \omega_0$, i.e. the identity map is a conformal symplectomorphism $\mathrm{id} : (M, \omega_0) \to (M, \omega_1)$. It is important to note that for us the form ω is the structure not its conformal equivalence class, as for some authors. In other words conformally equivalent lcs forms on a given manifold determine distinct but isomorphic objects of the category, whose objects are lcs manifolds and morphisms conformal symplectomorphisms.

Example 2.9. One example of a first kind lcs pair is carried by the mapping torus of a strict contactomorphism, see Banyaga [2]. The mapping torus $M_{\phi,c} := C \times \mathbb{R}/\sim$, $(x, \theta) \sim (\phi(x), \theta + 1)$, of a strict contactomorphism ϕ of (C, λ) fibers over S^1 with fiber C,

$$\pi : M_{\phi,c} \to S^1,$$

and carries the lcs pair (λ_ϕ, α), where λ_ϕ is induced by λ, well defined since $\phi^*\lambda = \lambda$, with $\alpha = c\pi^*(d\theta)$, for the parameter $0 \neq c \in \mathbb{R}$. In particular, these are scale integral first kind lcs pairs.

Moreover we have:

Theorem 2.10 (Only reformulating Bazzoni–Marrero [3, Theorem 4.7])**.** *A first kind lcs pair (λ, α) on a closed connected manifold M of dimension at least 4 is Liouville isomorphic to the mapping torus of a strict contactomorphism if and only if it is scale integral.*

Bazzoni–Marrero assume M connected and that the foliation $\ker \alpha$ has a compact leaf; for a first kind pair on closed M the latter holds when α is scale integral, the leaves being the components of the fibers of the map p of Proposition 1.18, which applies as $\alpha(X_\lambda) = -\lambda(X_\alpha) = -1$. The (scale) integrality condition is of course necessary since the Lee form of a mapping torus of a strict contactomorphism will have this property. Thus we may understand scale irrational first kind lcs pairs as first (and rather dramatic) departures from the world of contact manifolds into a brave new lcs world.

Remark 2.11. Note that scale irrational first kind lcs pairs certainly exist. A simple example is given by taking closed 1-forms α, λ on T^2 with α scale irrational and $\lambda \wedge \alpha$ nowhere vanishing, for instance $\alpha = dy + s\,dx$ with s irrational and $\lambda = dx$. Then (λ, α) is a scale irrational first kind lcs pair on T^2, with lcs form $\lambda \wedge \alpha$ and $\lambda(X_\alpha) = 1$. It is not Liouville isomorphic to the mapping torus of a strict contactomorphism: a Liouville isomorphism pulls back the closed 1-form of the pair, and that of the mapping torus, $c\,\pi^*d\theta$, is scale integral. In general, on a closed manifold M with $b_1(M) \geq 2$ we may perturb a (first kind) scale integral lcs pair to a (first kind) scale irrational one. When $b_1(M) = 1$ every closed 1-form is scale integral, see the remarks following Corollary 1.19. The examples of the present paper deal with deformations of this sort.

3. On Reeb 2-curves

Definition 3.1. Define ***the set*** $\mathcal{L}(M)$ ***of lcs pairs on*** M, to be:

$$\mathcal{L}(M) = \{(\lambda, \alpha) \in \Omega^1(M) \times \Omega^1(M) \,|\, \alpha \text{ is closed, } d_\alpha \lambda \text{ is non-degenerate}\}.$$

Define $\mathcal{F}(M) \subset \mathcal{L}(M)$ to be the subset of first kind lcs pairs.

In what follows we use the following metric on $\mathcal{L}(M)$, which induces the C^∞ topology. For (λ_1, α_1), (λ_2, α_2) in $\mathcal{L}(M)$ set

$$d_\infty((\lambda_1, \alpha_1), (\lambda_2, \alpha_2)) = d_{C^\infty}(\lambda_1, \lambda_2) + d_{C^\infty}(\alpha_1, \alpha_2), \tag{3.2}$$

where d_{C^∞} on the right side is the usual C^∞ metric.

Theorem 3.3 below is proved in Section 7, by counting elliptic pseudo-holomorphic curves in M; Theorem 3.4 is proved there by an elementary argument.

Theorem 3.3. *Let (C, λ) be a closed contact manifold, satisfying one of the following conditions:*

1. *(C, λ) has at least one non-degenerate closed Reeb orbit.*
2. *$i(N, R^\lambda, \beta) \neq 0$ where the latter is the Fuller index of some open compact subset of the orbit space: $N \subset \mathcal{O}(R^\lambda, \beta)$, as previously.*

Then we have the following:

1. *For some d_∞ neighborhood U of the lcs-fication (λ, α) in the space $\mathcal{F}(M = C \times S^1)$, every element of U admits a Reeb 2-curve.*
2. *For any $(\lambda', \alpha') \in U$, the corresponding Reeb 2-curve $u : \Sigma \to M$ can be assumed to be* **elliptic** *meaning that Σ is elliptic (more specifically: a nodal elliptic curve, see Section 4.1).*
3. *If α' is integral, so that charge is defined, then u can also be assumed to be α'-charge 1 (see Definition 4.7).*
4. *If M has dimension 4 then u can be assumed to be embedded and normal (the set of nodes is empty). And so in particular, such a u represents a closed, $(\omega' = d_{\alpha'}\lambda')$-symplectic embedded 2-torus in M, tangent to the canonical distribution of (λ', α').*

Theorem 3.4. *Suppose that (λ, α) is a Reeb pair on M. If M has an immersed Reeb 2-curve, Definition 4.2, then it also has a Reeb 1-curve. Furthermore, if it has an immersed elliptic Reeb 2-curve, then this Reeb 2-curve is normal.*

We have an immediate corollary of Theorem 3.3 and Theorem 3.4, using that elements of $\mathcal{F}(M)$ satisfy the Reeb condition, as $\lambda(X_\alpha) = 1$ by the computation following Definition 2.6, and that, as $\dim M = 4$, the Reeb 2-curve of the fourth part of Theorem 3.3 is embedded, hence immersed, by Definition 4.2.

Corollary 3.5. *Let λ be a contact form on a closed 3-manifold C, with at least one non-degenerate closed Reeb orbit, or more generally satisfying $i(N, R^\lambda, \beta) \neq 0$ for some open compact N as previously. Then there is a d_∞ neighborhood U of the lcs-fication (λ, α) in the space $\mathcal{F}(C \times S^1)$, s.t. for each $(\lambda', \alpha') \in U$ there is a Reeb 1-curve.*

Lemma 3.6. *Let (λ, α) be an lcs pair on a closed manifold M. Then $0 \neq [\alpha] \in H^1(M, \mathbb{R})$.*

Proof. Suppose by contradiction that α is exact and let g be a primitive. Then computing we get: $d_\alpha \lambda = \frac{1}{f} d(f\lambda)$ with $f = e^{-g}$. Consequently, $d(f\lambda)$ is non-degenerate on M which contradicts Stokes theorem. □

Proof of Proposition 1.16. Let (λ, α) be a Reeb pair on a closed surface M. Then $\omega(X_\lambda, X_\alpha) = \lambda(X_\alpha) > 0$, so X_λ, X_α are linearly independent and $\mathcal{D} = TM$. Hence the identity map $M \to M$, for any complex structure on the surface M, satisfies the first condition of Definition 1.12, and the second by Lemma 3.6. So it is an immersed Reeb 2-curve, and the claim follows by Theorem 3.4. □

4. J-holomorphic curves in certain lcs manifolds

Let M, J be an almost complex manifold and Σ, j a Riemann surface. Recall that a map $u : \Sigma \to M$ is said to be J-*holomorphic* if $du \circ j = J \circ du$.

Notation 4.1. We will often say ***J-curve*** in place of J-holomorphic curve.

To set notation and terminology we review the basic definition of a nodal curve.

Definition 4.2. A ***nodal Riemann surface*** (without boundary) is a pair $\Sigma = (\widetilde{\Sigma}, \mathcal{N})$ where $\widetilde{\Sigma}$ is a Riemann surface, and $\mathcal{N}$ a set of pairs of points of $\widetilde{\Sigma}$, possibly empty: $\mathcal{N} = \{(z_0^0, z_0^1), \dots, (z_n^0, z_n^1)\}$, with $z_i^j \neq z_k^l$ for $(i,j) \neq (k,l)$. By slight abuse, we may also denote by Σ the quotient space $\widetilde{\Sigma}/\sim$, where the equivalence relation is generated by $z_i^0 \sim z_i^1$. Let $q_\Sigma : \widetilde{\Sigma} \to (\widetilde{\Sigma}/\sim)$ denote the quotient map. The elements $q_\Sigma(\{z_i^0, z_i^1\}) \in \widetilde{\Sigma}/\sim$, are called ***nodes***. Let M be a smooth manifold. By a map $u : \Sigma \to M$ of a nodal Riemann surface Σ, we mean a set map $u : (\widetilde{\Sigma}/\sim) \to M$. u is called smooth or immersion or J-holomorphic (when M is almost complex) if the map $\widetilde{u} = u \circ q_\Sigma$ is smooth or respectively immersion or respectively J-holomorphic. We call $\widetilde{u}$ the ***normalization of*** u. u is called an embedding if u is a topological embedding and its normalization is an immersion. The cohomology groups of Σ are defined as $H^\bullet(\Sigma) := H^\bullet(\widetilde{\Sigma}/\sim)$, likewise with homology. The genus of Σ, for Σ connected, is its arithmetic genus $\sum_i g(\Sigma_i) + |\mathcal{N}| - c + 1$, for Σ_i the components of $\widetilde{\Sigma}$ and c their number; the nodal surfaces of genus one are described in Section 4.1.

We shall say that $(\widetilde{\Sigma}, \mathcal{N})$ is normal if $\mathcal{N} = \emptyset$. Similarly, $u : \Sigma \to M$, $\Sigma = (\widetilde{\Sigma}, \mathcal{N})$ is called ***normal*** if $\mathcal{N} = \emptyset$. Note that if u is a Reeb 2-curve, its normalization $\widetilde{u}$ may not be a Reeb 2-curve (the second condition may fail).

Tamed first kind lcs pairs, Example 4.4 below, give immediate examples of almost complex manifolds where the L^2 energy functional is unbounded on the moduli spaces of fixed homology class J-curves, as well as where null-homologous J-curves can be non-constant. We are going to see this shortly after developing a more general theory.

Definition 4.3. Let (λ, α) be a Reeb pair on M, Definition 1.10, that is with $\omega(X_\lambda, X_\alpha) = \lambda(X_\alpha) > 0$, where $\omega = d_\alpha \lambda$. In this case, $\mathcal{D}$ is a 2-dimensional distribution, and we say that an ω-compatible J, that is one with $\omega(\cdot, J\cdot)$ a Riemannian metric, is (λ, α)***-admissible*** if J preserves the canonical distribution $\mathcal{D}$ and its ω-orthogonal complement $\mathcal{D}^\perp$: $J(\mathcal{D}) \subset \mathcal{D}$ and $J(\mathcal{D}^\perp) \subset \mathcal{D}^\perp$. Then $d\lambda$ tames J on $\mathcal{D}^\perp$, that is $d\lambda(v, Jv) > 0$ for $0 \neq v \in \mathcal{D}^\perp$, automatically, since $\mathcal{D}^\perp = \ker \alpha \cap \ker \lambda$, on which $\omega = d\lambda$. We call (λ, α, J) as above a ***tamed lcs pair***.

Admissible J exist, and the space of such J is contractible: they are the compatible structures on the symplectic subbundles $\mathcal{D}$ and $\mathcal{D}^\perp$ taken separately, and each such space is contractible, cf. [27, Proposition 2.50].

Example 4.4. If (λ, α) is a first kind lcs pair on M then by the computation following Definition 2.6, $\omega(X_\lambda, X_\alpha) = 1$ everywhere. In particular, we may find a J such that (λ, α, J) is a tamed lcs pair, and the space of such J is contractible. We call such (λ, α, J) a ***tamed first kind lcs pair***.

Lemma 4.5. *Let (λ, α, J) be a tamed first kind lcs pair on M. Then given a smooth $u : \Sigma \to M$, where Σ is a closed (nodal) Riemann surface, u is J-holomorphic only if*

$$\operatorname{image} d\widetilde{u}(z) \subset \mathcal{V}_\lambda(\widetilde{u}(z))$$

for all $z \in \widetilde{\Sigma}$, where $\widetilde{u} : \widetilde{\Sigma} \to M$ is the normalization of u (see Definition 4.2). In particular $\widetilde{u}^* d\lambda = 0$.

Proof. By the first kind condition (see the computation following Definition 2.6) we have

$$V := \mathcal{V}_\lambda = \mathcal{D}.$$

Let u be J-holomorphic, so that $\widetilde{u}$ is J-holomorphic (by definition of a J-holomorphic nodal map). We have

$$\int_{\widetilde{\Sigma}} \widetilde{u}^* d\lambda = 0$$

by Stokes theorem. Let $V^\perp := \mathcal{D}^\perp$ be the ω-orthogonal complement, as in the definition of admissibility, and $\operatorname{proj}(p) : T_pM \to V^\perp(p)$ the projection induced by the splitting $TM = V \oplus V^\perp$.

Write $d\widetilde{u}^\perp := \operatorname{proj} \circ d\widetilde{u}$. As J preserves the splitting $TM = V \oplus V^\perp$, $d\widetilde{u}^\perp \circ j = J \circ d\widetilde{u}^\perp$. As $\iota_v d\lambda = 0$ for $v \in V$, by definition of $\mathcal{V}_\lambda$, for $a \in T_z\widetilde{\Sigma}$ we get

$$\widetilde{u}^* d\lambda(a, ja) = d\lambda(d\widetilde{u}^\perp(a), Jd\widetilde{u}^\perp(a)) \geq 0,$$

with equality iff $d\widetilde{u}^\perp(a) = 0$, as $d\lambda$ tames J on $V^\perp$. So $\widetilde{u}^* d\lambda$ is a non-negative multiple of a positive 2-form on $\widetilde{\Sigma}$, positive on the open set $\{\operatorname{proj} \circ d\widetilde{u} \neq 0\}$. If this set is non-empty, $\int_{\widetilde{\Sigma}} \widetilde{u}^* d\lambda > 0$, a contradiction. Thus,

$$\forall z \in \widetilde{\Sigma} : \operatorname{proj} \circ d\widetilde{u}(z) = 0,$$

so

$$\forall z \in \widetilde{\Sigma} : \operatorname{image} d\widetilde{u}(z) \subset \mathcal{V}_\lambda(\widetilde{u}(z)).$$

□

Example 4.6. Let (λ, α) on $C \times S^1$ be the lcs-fication of a contact manifold (C, λ). In this case

$$X_\alpha = (R^\lambda, 0),$$

where R^λ is the Reeb vector field and

$$X_\lambda = (0, -\partial_\theta),$$

minus the vector field generating the natural action of S^1 on $C \times S^1$; the sign is forced by

$$\omega(\partial_\theta, \cdot) = -(\alpha \wedge \lambda)(\partial_\theta, \cdot) = -\lambda.$$

If we denote by $\xi \subset T(C \times S^1)$ the distribution $\xi := \ker \lambda \cap \ker \alpha = \{(v, 0) \mid v \in \ker \lambda\}$, the horizontal lift of the contact distribution of C, then in this case $\xi = \mathcal{D}^\perp$, the ω-orthogonal complement of the canonical distribution, by Definition 4.3. We then take J to be an almost complex structure on ξ, which is S^1 invariant, and compatible with $d\lambda$. The latter means that

$$g_J(\cdot, \cdot) := d\lambda|_\xi(\cdot, J\cdot)$$

is a J invariant Riemannian metric on the distribution ξ.

There is an induced almost complex structure J^λ on $C \times S^1$, which is S^1-invariant, coincides with J on ξ and which satisfies:

$$J^\lambda(X_\alpha) = -X_\lambda, \quad \text{that is} \quad J^\lambda(R^\lambda, 0) = (0, \partial_\theta).$$

It is ω-compatible, as $\omega = d\lambda$ on ξ, $\omega(X_\alpha, J^\lambda X_\alpha) = \omega((R^\lambda, 0), (0, \partial_\theta)) = \lambda(R^\lambda) = 1$; and it preserves ξ and $\mathcal{D}$. Then $(\lambda, \alpha, J^\lambda)$ is a tamed integral first kind lcs pair on $C \times S^1$.

4.1. **Charged elliptic curves in an lcs manifold.** We now study moduli spaces of elliptic curves in an lcs manifold, constrained to have a certain charge.[1] In the present context, one reason for the introduction of "charge" is that it is now possible for non-constant holomorphic curves to be null-homologous, so we need additional control. Here is a simple example: take $S^3 \times S^1$ with $J = J^\lambda$, for λ the standard contact form; then all the Reeb tori (Section 4.2) are null-homologous. Moreover the Reeb torus of a Reeb orbit of period c has $\int_{T^2} u_o^* \omega = c$, as $u_o^* d\lambda = 0$ and $u_o^*(\alpha \wedge \lambda) = -c\, ds \wedge dt$. This is half its L^2 energy for the metric $\omega(\cdot, J^\lambda \cdot)$, as $|du|^2 \operatorname{dvol} = 2u^*\omega$ for a J^λ-holomorphic u, [28, 2nd ed., Lemma 2.2.1]. So the energy is unbounded on the iterates of a Hopf fiber, all of class 0.

By an *elliptic curve* we mean a closed connected Riemann surface of genus one, that is a complex torus, and by a *nodal elliptic curve* a connected nodal Riemann surface of genus one, see Definition 4.2: a complex torus with trees of spheres attached, or a cycle of spheres, of length one (a torus with a pinched cycle) or more, with trees of spheres attached. We use no marked points.

The notion of charge can be defined in a general setting. Let M be a manifold endowed with a closed integral 1-form α. Let $u : T^2 \to M$ be a continuous map with $0 \neq u^*[\alpha] \in H^1(T^2, \mathbb{Z})$, and let $g : T^2 \to S^1$ be a fibration with $g^*[d\theta] = u^*[\alpha]$, for instance a linear map $T^2 = \mathbb{R}^2/\mathbb{Z}^2 \to \mathbb{R}/\mathbb{Z}$ in that class. Take a component of a fiber of g, an embedded circle, and let $[\gamma] \in H_1(T^2, \mathbb{Z})$ be its class for one of its two orientations, represented by $\gamma : S^1 \to T^2$. Let $[\rho] \in H_1(T^2, \mathbb{Z})$, likewise represented by $\rho : S^1 \to T^2$, be a class with $[\gamma] \cdot [\rho] = 1$, where $\cdot$ is the intersection pairing for the orientation of T^2, the complex orientation when T^2 is a Riemann surface; it exists as $[\gamma]$ is primitive. Then

$$\langle [\gamma], u^*[\alpha] \rangle = 0, \quad \langle [\rho], u^*[\alpha] \rangle \neq 0,$$

where $\langle , \rangle$ is the natural pairing of homology and cohomology; the first as $g \circ \gamma$ is constant, the second since $[\gamma], [\rho]$ generate $H_1(T^2, \mathbb{Z})$. Reversing the orientation of γ replaces $[\rho]$ by $-[\rho]$ modulo $[\gamma]$ and changes the sign of $\langle [\rho], u^*[\alpha] \rangle$; we orient γ so that $\langle [\rho], u^*[\alpha] \rangle > 0$.

Definition 4.7. For u, γ, ρ as above we call

$$k = |\langle [\rho], u^*[\alpha] \rangle| \in \mathbb{N}_{>0},$$

the ***α-charge of*** u, or just the charge of u when α is implicit. The class $[u \circ \gamma] \in \pi_1(M)$ will be called the ***π-class of*** u, where here and below $\pi_1(M)$ denotes the set of free homotopy classes of loops in M.

It is easy to see that the α-charge and the π-class do not depend on the choices above. The π-class depends on the orientation of T^2: reversing it replaces the π-class by that of the inverse loop. When $u^*[\alpha] = 0$ we set the charge to be 0. We extend both to maps $u : \Sigma \to M$ with Σ a nodal elliptic curve. There are two cases, see the description of nodal elliptic curves above. If Σ has a component Σ_1 of genus one, a torus component, the charge and the π-class of u are those of $u|_{\Sigma_1}$. Otherwise Σ is a cycle of spheres with trees of spheres attached, and $H_1(\Sigma, \mathbb{Z}) \simeq \mathbb{Z}$. In this case the charge of u is $|\langle [\rho], u^*[\alpha] \rangle|$ for ρ a generator, and the π-class of u is defined to be the trivial class. This is the definition under which the π-class is locally constant in the C^0-topology of Lemma 4.9 below. The charge is preserved under Gromov convergence of stable maps, see Lemma 4.9 below. But it is not determined by the homology class, so that charge is *not* a functional $H_2(M, \mathbb{Z}) \to \mathbb{N}$.

Definition 4.8. By the above, associated to a continuous map $u : \Sigma \to M$ with Σ a nodal elliptic curve, and non-zero α-charge, we have a triple $(A = u_*[\Sigma], \beta, k) \in H_2(M, \mathbb{Z}) \times \pi_1(M) \times \mathbb{N}_{>0}$, corresponding to the homology class, for $[\Sigma] := q_{\Sigma *} \sum_i [\Sigma_i] \in H_2(\Sigma)$, the sum over the components Σ_i of $\widetilde{\Sigma}$, the π-class, and the α-charge. This triple will be called the ***charge class*** of u.

[1] The name charge is inspired by the notion of charge in Oh–Wang [31], in the context of contact instantons. However, the respective notions are not obviously related.

Let (M,J) be an almost complex manifold and α a closed integral 1-form on M non-vanishing in cohomology, then we call (M,J,α) a ***Lee manifold***. For $k\geq 1$ we define

$$\overline{\mathcal{M}}_1^k(J,A,\beta)$$

as the set of equivalence classes of pairs (u,Σ), for Σ a possibly nodal elliptic curve and $u:\Sigma\to M$ a J-holomorphic map of charge class (A,β,k), understood as a Kontsevich stable map when Σ has spherical components. The bar refers, as usual, to the closure by nodal curves; a priori the space is not compact. The equivalence relation is $(u_1,\Sigma')\sim(u_2,\Sigma'')$ if there is a biholomorphism $\phi:\Sigma'\to\Sigma''$ s.t. $u_2\circ\phi=u_1$. Here a biholomorphism of nodal Riemann surfaces $\Sigma'=(\widetilde{\Sigma}',\mathcal{N}')$, $\Sigma''=(\widetilde{\Sigma}'',\mathcal{N}'')$ is a biholomorphism $\widetilde{\phi}:\widetilde{\Sigma}'\to\widetilde{\Sigma}''$ taking $\mathcal{N}'$ onto $\mathcal{N}''$, as sets of unordered pairs; it descends to a homeomorphism ϕ of the quotients, and $u_2\circ\phi=u_1$ means $\widetilde{u}_2\circ\widetilde{\phi}=\widetilde{u}_1$. Such a ϕ preserves the charge class, so that $\overline{\mathcal{M}}_1^k(J,A,\beta)$ is well defined. There are no marked points, so the automorphism group of an element is its stabilizer in the automorphism group of the domain, which is finite: a non-constant J-holomorphic torus is invariant under no one parameter group of translations, as a J-holomorphic map whose differential has rank at most one everywhere is constant, and a stable map with spherical components has finitely many automorphisms by stability.

By slight abuse we denote such an equivalence class simply by u, so we may write $u\in\overline{\mathcal{M}}_1^k(J,A,\beta)$, with Σ implicit. The expected dimension of $\overline{\mathcal{M}}_1^k(J,A,\beta)$ is $2c_1(A)$: the Fredholm index of the linearized Cauchy–Riemann operator at u, coupled with the variations of the complex structure of the domain, cf. (6.8), which is $2c_1(A)+2$ by the Riemann–Roch theorem, minus the dimension of the translation group of the domain torus, which is 2. On the lcs-fication $C\times S^1$ of (C,λ), for the classes A_β of Section 6.1 one has $c_1(A_\beta)=0$ for every J homotopic to J^λ, for instance every (λ,α)-admissible J, the space of these being contractible and containing J^λ: for J^λ, $TM=\mathcal{D}\oplus\xi$ as complex vector bundles, with $\mathcal{D}$ trivial and ξ the pullback of $\ker\lambda$ from C, so $c_1(TM)=pr_C^*c_1(\ker\lambda)$ pairs to zero with $A_\beta=\beta\otimes[S^1]$; so the expected dimension is 0.

Lemma 4.9. *Let M be a manifold with a closed integral 1-form α, and let $u_i:\Sigma_i\to M$ be continuous stable maps of nodal elliptic curves (stable meaning that constant components are stable curves, as in Section 5) converging in the C^0-topology of Siebert* [40, Definition 3.6] *to $u:\Sigma\to M$. Then for i large $u_i^*[\alpha]\neq 0$ iff $u^*[\alpha]\neq 0$, and in this case u_i and u have the same charge class. In particular, for an almost complex structure J on M, $\overline{\mathcal{M}}_1^k(J,A,\beta)$ is open and closed in the space of J-holomorphic stable maps of nodal elliptic curves in class A, with the Gromov topology of Section 5.*

Proof. Siebert's C^0-topology is defined for all continuous stable maps, and on J-holomorphic stable maps it coincides with the Gromov topology, [40, Remark 3.12]; the conclusions, about convergent sequences and about open and closed sets, persist in any finer topology, in particular in that of Section 5. By its definition, [40, Definitions 2.13, 3.6], for i large there is a continuous surjection $\varphi_i:\Sigma_i\to\Sigma$ with u_i uniformly close to $u\circ\varphi_i$, hence homotopic to it; φ_i collapses finitely many disjoint embedded circles in Σ_i to nodes of Σ and is an orientation preserving diffeomorphism elsewhere. Write H_1 for integral first homology. Collapsing a circle is, up to homotopy equivalence, attaching a disc along it, so $\varphi_{i*}:H_1(\Sigma_i)\to H_1(\Sigma)$ is onto. Also $\varphi_{i*}[\Sigma_i]=[\Sigma]$, so $u_{i*}[\Sigma_i]=u_*[\Sigma]$: the homology classes agree.

The charge of a map $v:\Sigma'\to M$ of a nodal elliptic curve is the non-negative generator of the subgroup $\langle v_*H_1(\Sigma'),[\alpha]\rangle$ of $\mathbb{Z}$, zero iff $v^*[\alpha]=0$; when Σ' has a torus component Σ'_1 this uses $H_1(\Sigma')=H_1(\Sigma'_1)$, the other components being spheres. As $u_{i*}=u_*\varphi_{i*}$ on H_1 and φ_{i*} is onto, u_i and u have the same charge.

Suppose the charge k is positive. If Σ_i is a cycle of spheres then so is Σ, as φ_{i*} maps onto $H_1(\Sigma)$, and the π-classes of u_i and u are both trivial by definition. Otherwise let $\gamma_i:S^1\to\Sigma_i$ be the oriented fiber circle of Definition 4.7 on the torus component of Σ_i. The π-class of u_i is $[u_i\circ\gamma_i]=[u\circ\varphi_i\circ\gamma_i]$, and $\langle u^*[\alpha],\varphi_{i*}[\gamma_i]\rangle=\langle u_i^*[\alpha],[\gamma_i]\rangle=0$. If Σ is a cycle of spheres then $\pi_1(\Sigma)=H_1(\Sigma)=\mathbb{Z}$, on which

$\langle u^*[\alpha], \cdot\rangle$ is injective, so $\varphi_i \circ \gamma_i$ is null-homotopic, and the π-class of u_i is trivial, as is that of u by definition. If Σ has a torus component Σ_1 then so does Σ_i, as φ_{i*} maps onto $H_1(\Sigma) \simeq \mathbb{Z}^2$, and φ_{i*} is an isomorphism on H_1. It preserves the intersection pairing, which on $H_1(\Sigma)$ is that of Σ_1: composed with the collapse of the sphere trees of Σ to their attaching points, φ_i restricts to a degree one map between the torus components, being an orientation preserving homeomorphism over the complement of finitely many points, and a degree one map between tori induces an element of $SL_2(\mathbb{Z})$ on H_1. So $\varphi_{i*}[\gamma_i]$ is primitive, pairs to zero with $u^*[\alpha]$, and has intersection number one with $\varphi_{i*}[\rho_i]$, which pairs to $k > 0$ with $u^*[\alpha]$, for ρ_i as in Definition 4.7: it is the fiber class $[\gamma]$ of u. As $\pi_1(\Sigma)$ is abelian, $\varphi_i \circ \gamma_i$ is freely homotopic to γ, and the π-classes agree.

Finally, by [40, Definition 3.6] the above applies to every u' in a basic neighborhood of u in place of the u_i, so the charge class is locally constant, and $\overline{\mathcal{M}}_1^k(J, A, \beta)$ is the preimage of (A, β, k) under it, on the open and closed set of positive charge. □

4.2. **Reeb holomorphic tori in $(C \times S^1, J^\lambda)$.** In this subsection we discuss an important example. Let (C, λ) be a contact manifold and let α and J^λ be as in Example 4.6. In particular $(C \times S^1, J^\lambda, \alpha)$ is a Lee manifold.

In this case we have one natural type of charge 1 J^λ-holomorphic tori in $M = C \times S^1$. Let o be a period c closed Reeb orbit of R^λ, and let β be its class in $\pi_1(C) \subset \pi_1(M)$. A ***Reeb torus*** u_o for o is the map

$$u_o : (S^1 \times S^1 = T^2) \to C \times S^1$$
$$u_o(s, t) = (o(s), t).$$

A Reeb torus is J^λ-holomorphic for a uniquely determined complex structure j on T^2 defined by:

$$j(\frac{\partial}{\partial s}) = c\frac{\partial}{\partial t}.$$

We have $u_o^*\alpha = dt$, so u_o has charge 1. Its fiber class is that of $S^1 \times \{t\}$ oriented by ∂_s, as (∂_s, ∂_t) is the complex orientation and $\langle [\{s\} \times S^1], dt\rangle = 1 > 0$; so its π-class is $[o] = \beta$, and $u_{o*}[T^2] = \beta \otimes [S^1]$, the class A_β of Section 6.1.

5. Elements of Gromov–Witten theory of an almost complex manifold

Throughout we suppose that (M, J) is a compact almost complex manifold, where the almost complex structures J are assumed in the paper to be C^∞.

We first recall the moduli spaces, and the energy quantization theorem. Let $\mathcal{M}_{g,n}(J, A)$ denote the moduli space of isomorphism classes of class A, J-holomorphic curves in M, with domain a connected genus g closed Riemann surface, with n marked labeled points. Here an isomorphism between $u_1 : \Sigma_1 \to M$ and $u_2 : \Sigma_2 \to M$ is a biholomorphism of marked Riemann surfaces $\phi : \Sigma_1 \to \Sigma_2$ s.t. $u_2 \circ \phi = u_1$.

Theorem 5.1 ([28, Proposition 4.1.4])**.** *Let (M, J) be a compact almost complex manifold, and $u : (S^2, j) \to M$ a J-holomorphic map. Given a Riemannian metric g on M, there is an $\hbar = \hbar(g, J) > 0$ s.t. if $e_g(u) < \hbar$ then u is constant, where e_g is the L^2-energy functional,*

$$e_g(u) = \int_{S^2} |du|^2 \,\mathrm{dvol}\,.$$

Next we state the basic precompactness theorem.

Theorem 5.2. *Let (M, J) be a compact almost complex manifold. Then $\mathcal{M}_{g,n}(J, A)$ is an open subset of the space*

$$\overline{\mathcal{M}}_{g,n}(J, A),$$

of Kontsevich stable maps, with the natural Gromov topology, see for instance [28, Section 5.6]*, for the genus 0 case, and* [32, Section 9.1] *for the general case. Moreover, given $E > 0$, the subspace $\overline{\mathcal{M}}_{g,n}(J,A)_E \subset \overline{\mathcal{M}}_{g,n}(J,A)$ consisting of elements u with $e(u) \leq E$ is compact, where e is the L^2 energy with respect to an auxiliary J-invariant metric. In other words e is a proper function, continuous as Gromov convergence includes convergence of energy. Here Gromov convergence of stable maps, also for varying J_j, is the sequential notion of* [28, 2nd ed., Definition 5.2.1] *and* [17]*: the domains converge in a Deligne–Mumford space after adding marked points, the maps converge in C^∞ away from the nodes after reparametrization, and the energies converge; in particular the maps converge uniformly after reparametrization, in the C^0-topology of Siebert used in Lemma 4.9. More generally, if $J_j \to J$ in C^∞ and u^j are J_j-holomorphic stable maps in class A with $e(u^j) \leq E$, then a subsequence Gromov converges to a J-holomorphic stable map. For closed curves of any genus in a compact almost complex manifold, with the complex structures of the domains allowed to degenerate, this is proved in Hummel* [17].

Proof of Theorem 5.2. (Outline, as the argument is standard.) Suppose that we have a sequence u^j of J-holomorphic maps with L^2-energy $\leq E$. By [28, Theorem 4.1.1], a sequence u^j of J-holomorphic curves has a convergent subsequence if $\sup_j ||du^j||_{L^\infty} < \infty$. On the other hand, when this condition does not hold, a rescaling argument tells us that a holomorphic sphere bubbles off. The quantization Theorem 5.1, tells us that the energy of a non-constant J-holomorphic map of $\mathbb{CP}^1$ is at least $\hbar > 0$. So if the energy of the maps u^j is bounded from above by E, only finitely many bubbles may appear, so that a subsequence of u^j must converge in the Gromov topology to a Kontsevich stable map. When the complex structures of the domains degenerate, the same bubbling analysis applies on the long necks, whose thin parts carry little energy and become the nodes of the limit; the details, for any genus and for varying J_j, are in [17]. □

We now specialize to genus 1 and no marked points, the only case used in this paper. Let (M, J, α) be a Lee manifold, and let $\mathcal{M} := \overline{\mathcal{M}}_1^k(J, A, \beta)$ be the moduli space of Section 4.1. It is an open and closed subset of $\overline{\mathcal{M}}_{1,0}(J,A)$ by Lemma 4.9, so Theorem 5.2 applies to it; and it contains no constant maps, as $k \geq 1$. Its Gromov topology is the topology of [32, Section 9.1], there described by neighborhood bases of maps whose graphs over domains, stabilized by extra marked points, are Hausdorff-close; this topology is first countable, and by the following lemma, which uses no taming form, its convergent sequences are the Gromov convergent ones.

Lemma 5.3. *Let $u_i : \Sigma_i \to M$ be J-holomorphic stable maps converging to a J-holomorphic stable map $u_0 : \Sigma_0 \to M$ in the topology of* [32, Section 9.1]*: the domains converge in a Deligne–Mumford space after adding marked points, and u_i is uniformly δ_i-close to $u_0 \circ \varphi_i$, $\delta_i \to 0$, for collapse maps $\varphi_i : \Sigma_i \to \Sigma_0$. Then the energies $e(u_i)$ are bounded, and u_i Gromov converges to u_0.*

Proof. Let $\omega_g := g(J\cdot, \cdot)$ for the auxiliary J-invariant metric g. For a J-holomorphic map u we have $|du|^2 \operatorname{dvol} = 2u^*\omega_g$, [28, 2nd ed., Lemma 2.2.1], so $e(u_i) = 2\int u_i^*\omega_g$. Let $H_i : \Sigma_i \times [0,1] \to M$ be the geodesic homotopy from $u_0 \circ \varphi_i$ to u_i. By Stokes, $\int u_i^*\omega_g - \int (u_0 \circ \varphi_i)^*\omega_g = \int H_i^* d\omega_g$, and pointwise $|H_i^* d\omega_g| \leq C\delta_i(|du_i|^2 + |d(u_0 \circ \varphi_i)|^2)$, for C depending on g and $d\omega_g$. Now $\int (u_0 \circ \varphi_i)^*\omega_g = \int u_0^*\omega_g$, as φ_i has degree one, and $\int |d(u_0 \circ \varphi_i)|^2 \leq e(u_0) + \delta_i$ for φ_i chosen conformal outside thin annuli around the collapsed circles, mapped into small discs around the nodes, which changes $u_0 \circ \varphi_i$ little. So $(1 - 2C\delta_i)e(u_i) \leq (1 + 2C\delta_i)e(u_0) + 2C\delta_i^2$, and the energies are bounded. By Theorem 5.2 a subsequence Gromov converges; its limit is u_0, as Gromov convergence implies convergence in the topology of [32, Section 9.1], which is the C^0-topology of Siebert on J-holomorphic maps and is Hausdorff, [40, Proposition 3.8, Remark 3.12]. So u_i Gromov converges to u_0; cf. [40, Remark 3.12], stated there for any almost complex manifold with a Riemannian metric. □

Let $N \subset \mathcal{M}$ be an open compact subset, not necessarily connected. In our applications such N appear from concrete geometry, for example they may correspond to components of the space of Reeb strings.

Compactness is thus assumed, and is not emergent, and there is no general theory even within lcs geometry where it emerges.

We shall primarily refer in what follows to work of Pardon in [32], being more familiar to the author. However, the original setup of Fukaya–Ono [12], and later expanded with Oh–Ohta [11] can also work here in the same fashion.

5.1. **Implicit atlases.** The construction in [32] of an implicit atlas on a moduli space of J-curves asks for J to be tamed by a symplectic form, but uses this only for compactness. We sketch it, following [32, Section 9.2], to locate that use, and to see that it applies to an open compact $N \subset \mathcal{M}$ as above.

Let Σ denote the domain of a curve u, a possibly nodal elliptic curve. Pardon's implicit atlas is indexed by the set $\mathfrak{A}$ of thickening datums, [32, Definition 9.2.1]. A datum $a \in \mathfrak{A}$ is a tuple $(Z_a, r_a, \Gamma_a, \overline{M}_a, E_a, \Theta_a)$ (we rename Pardon's α, D_α, $\overline{\mathcal{M}}_\alpha$, λ_α, ϕ_α, e_α to a, Z_a, $\overline{M}_a$, Θ_a, ψ_a, ε_a, to avoid clashes with our notation), consisting of: a compact codimension two submanifold with boundary $Z_a \subset M$; an integer $r_a \geq 1$; a finite group Γ_a; a smooth Γ_a-manifold $\overline{M}_a$, Γ_a acting possibly non-effectively, with an isomorphism of orbifolds of $\overline{M}_a/\Gamma_a$ onto an open subset of $\overline{M}_{1,r_a}/S_{r_a}$, for $\overline{M}_{1,r_a}$ the Deligne–Mumford space of stable genus one curves with r_a marked points and S_{r_a} the symmetric group, with the universal curve $\overline{\mathcal{C}}_a \to \overline{M}_a$ pulled back as an orbifold fiber product; a finite dimensional Γ_a-representation E_a; and a Γ_a-equivariant linear map $\Theta_a : E_a \to C^\infty(\overline{\mathcal{C}}_a \times M, \Omega^{0,1}_{\overline{\mathcal{C}}_a/\overline{M}_a} \otimes_{\mathbb{C}} TM)$, supported away from the nodes and marked points of the fibers of $\overline{\mathcal{C}}_a$.

For a finite $I \subset \mathfrak{A}$ the thickened moduli space $\mathcal{M}_I$ is the set of isomorphism classes of tuples $(u, \{\psi_a\}_{a\in I}, \{\varepsilon_a\}_{a\in I})$, where $u : \Sigma \to M$ is a smooth stable map of charge class (A, β, k), transverse to each Z_a, $a \in I$, with exactly r_a intersection points, away from the nodes and from ∂Z_a, stable meaning that constant components are stable curves, $\psi_a : \Sigma \to \overline{\mathcal{C}}_a$ is an isomorphism of Σ, with the points $u^{-1}(Z_a)$ added as marked points, onto a fiber of $\overline{\mathcal{C}}_a \to \overline{M}_a$, $\varepsilon_a \in E_a$, and

$$\overline{\partial}_J u + \sum_{a\in I} \Theta_a(\varepsilon_a)(\psi_a, u) = 0, \tag{5.4}$$

where $\Theta_a(\varepsilon_a)(\psi_a, u)$, the pullback of $\Theta_a(\varepsilon_a)$ by $(\psi_a, u) : \Sigma \to \overline{\mathcal{C}}_a \times M$, is a $(0,1)$-form on Σ with values in u^*TM. $\mathcal{M}_I$ has the Gromov topology, which for $I \neq \emptyset$ is by definition the topology of Hausdorff distance on the images $((\psi_a)_{a\in I}, u)(\Sigma) \subset \prod_{a\in I} \overline{\mathcal{C}}_a \times M$, together with the ε_a, [32, Definition 9.2.4, footnote 34]; as the ψ_a are injective this image determines the tuple up to isomorphism, so this is a metric topology; the group $\Gamma_I = \prod_{a\in I} \Gamma_a$ acts on it through the ψ_a and the ε_a, and $E_I = \bigoplus_{a\in I} E_a$. The Kuranishi map is $s_I(u, \psi, \varepsilon) = \varepsilon$. Its zero set, $\{\varepsilon = 0\}$, consists of tuples with $u \in \mathcal{M}$, and $s_I^{-1}(0)/\Gamma_I$ is the open subset of $\mathcal{M}$ of curves transverse to the Z_a, $a \in I$, with r_a intersection points, whose domains with these points added are isomorphic to fibers of the $\overline{\mathcal{C}}_a$: the footprint of $\mathcal{M}_I$. The regular locus $\mathcal{M}_I^{reg} \subset \mathcal{M}_I$ consists of the tuples with trivial automorphism group, automatic for $I \neq \emptyset$ as the ψ_a are injective, at which the linearization of (5.4) in (u, ε), for the fixed domain Σ, is onto. This is Pardon's atlas on $\mathcal{M}$, when $\mathcal{M}$ is compact. Here $\mathcal{M}_I$ is open and closed in Pardon's thickened space of all genus one stable maps in class A: convergence in the topology above implies convergence in the C^0-topology of Lemma 4.9, so the charge class is locally constant on it. For our N we set

$$N_I := \{(u, \psi, \varepsilon) \in \mathcal{M}_I : \varepsilon \neq 0 \text{ or } u \in N\},$$

discarding the honest solutions outside N, and $N_I^{reg} := N_I \cap \mathcal{M}_I^{reg}$. This is where the openness of N in $\mathcal{M}$ is used: $Z_{N,I} := \{\varepsilon = 0,\ u \in N\}$ is open in the closed set $s_I^{-1}(0)$, as the forgetful map $s_I^{-1}(0) \to \mathcal{M}$ is continuous by Lemma 5.3, so $s_I^{-1}(0) \setminus Z_{N,I}$ is closed in $\mathcal{M}_I$, and N_I is open in $\mathcal{M}_I$, with $s_I^{-1}(0) \cap N_I = Z_{N,I}$ and $N_\emptyset = N$. For a compact N not open in $\mathcal{M}$, honest curves outside N would accumulate at points of $Z_{N,I}$, which would then have no neighborhood in N_I that is a manifold, and the openness and submersion axioms below would fail. The virtual fundamental class is extracted from the system $\{(N_I, E_I, s_I, N_I^{reg})\}_I$, the *$N$-system*, by homological algebra, [32, Sections 4, 5].

Pardon's verification, [32, Theorem 9.2.10], of the axioms of an implicit atlas, [32, Definition 3.1.1], for the system $\{\mathcal{M}_I\}_I$ uses a taming form only through the compactness of $\mathcal{M}$. The compatibility and homeomorphism axioms, concerning the forgetful maps $s_{I'\setminus I}^{-1}(0) \to \mathcal{M}_I$, $I \subset I'$, are checked from the definitions, using Lemma 5.3 for $I = \emptyset$, where the target carries the Gromov topology. The openness and submersion axioms, that $\mathcal{M}_I^{reg}$ is open in $\mathcal{M}_I$ and a topological manifold of dimension $2c_1(A) + \dim E_I$, with $s_{I'\setminus I} : \mathcal{M}_{I'} \to E_{I'\setminus I}$ a topological submersion over the preimage of $\mathcal{M}_I^{reg}$ under the forgetful map, are local: they are proved by the implicit function theorem near points with smooth domain, and by gluing near points with nodal domain, [32, Proposition 9.2.6 and Appendix B]. The covering axiom, that every element of $\mathcal{M}$ lies in the image under the forgetful map of $s_I^{-1}(0) \cap \mathcal{M}_I^{reg}$ for some I, follows from domain stabilization, [32, Lemma 9.2.8], and from choosing E_a to surject onto the cokernel of the linearized $\overline{\partial}_J$ operator at the given curve. Hausdorffness of the thickenings, also required, holds for any J: for $I \neq \emptyset$ the topology is metric, hence Hausdorff, and for $I = \emptyset$ it is the Gromov topology of $\mathcal{M}$, Hausdorff in any genus for any almost complex structure, Siebert [40, Propositions 3.8, 3.11, Remark 3.12].

All these conditions are conditions on open subsets and on local models, so they restrict from $\{\mathcal{M}_I\}_I$ to the system $\{N_I\}_I$ of open subsets; for the homeomorphism axiom note that an element of $s_{I'\setminus I}^{-1}(0)$ and its image in $\mathcal{M}_I$ have the same u and the same ε_a, $a \in I$, the others being 0, so both lie in the N-system or neither does. So in the construction of the atlas the taming form is used in one place: [32, Definition 3.1.1] asks that the moduli space carrying the atlas be compact, which for $\mathcal{M}$ is Gromov compactness. In our situation the moduli space carrying the atlas is N, and its compactness is a hypothesis. So an open compact N as above carries a natural implicit atlas, hence a virtual fundamental class in the sense of [32], or a Kuranishi structure in the setup of [12].

5.2. **The Gromov–Witten invariant and its dependence.** We may thus define, when $c_1(A) = 0$, so that the virtual dimension $2c_1(A)$ of N vanishes,

$$GW_1^k(N, J, A, \beta) := \int_{[N]^{vir}} 1 \in \mathbb{Q}, \tag{5.5}$$

the degree of the virtual fundamental class of N, [32, Section 5.1, Definition 9.3.1].

We now study how these counts depend on N, J. Given a Fréchet smooth family $\{J_t\}$, $t \in [0,1]$, of almost complex structures on M, a smooth family for short, we denote by $\overline{\mathcal{M}}_1^k(\{J_t\}, A, \beta)$ the space of pairs (u,t), $u \in \overline{\mathcal{M}}_1^k(J_t, A, \beta)$, topologized as a subspace of $\mathcal{S}^k(A,\beta) \times [0,1]$, for $\mathcal{S}^k(A,\beta)$ the space of all smooth stable maps of charge class (A, β, k) with respect to α, defined before Lemma 5.10 using an auxiliary Riemannian metric on M; its subspace topology on J'-holomorphic curves is the Gromov topology for any J', see the paragraph following property (c) there, and the topology of the cobordism moduli space is likewise determined by its convergent sequences, by (c), (a), (b) and Theorem 5.2, so it does not depend on the metric.

Lemma 5.6. *Let $\{J_t\}$, $t \in [0,1]$, be a smooth family of almost complex structures on M, and α a closed integral 1-form on M. Suppose that $\widetilde{N}$ is an open compact subset of the cobordism moduli space $\overline{\mathcal{M}}_1^k(\{J_t\}, A, \beta)$, with $c_1(A) = 0$. Let*

$$N_i := \{u \,|\, (u,i) \in \widetilde{N}\}, \quad i = 0, 1,$$

open compact subsets of $\overline{\mathcal{M}}_1^k(J_i, A, \beta)$; then

$$GW_1^k(N_0, J_0, A, \beta) = GW_1^k(N_1, J_1, A, \beta).$$

In particular if $GW_1^k(N_0, J_0, A, \beta) \neq 0$, there is a J_1-holomorphic, stable, charge class (A, β, k) elliptic curve in M.

Proof of Lemma 5.6. This is the construction of Section 5.1, applied to the cobordism moduli space, where it goes through unchanged. Namely, we may construct exactly as in [32, Lemma 9.3.2] for the

proof that the Gromov–Witten invariants do not depend on J, a natural implicit atlas, with boundary in the sense of [32, Section 3.2], on $\widetilde{N}$, whose boundary is $N_0^{op} \sqcup N_1$ (op denoting opposite orientation). And so we immediately get

$$GW_1^k(N_0, J_0, A, \beta) = GW_1^k(N_1, J_1, A, \beta).$$

□

Remark 5.7. Positivity of the charge, $k \geq 1$, rules out degenerations to constant curves. This is not essential: on a closed manifold such degenerations are impossible by energy quantization for non-constant J-holomorphic curves of any genus, which follows from the monotonicity theorem, Zinger [48, Proposition 3.12], as noted to me by Spencer Cattalani.

We now state a basic technical lemma, following some standard definitions.

Definition 5.8. An ***almost symplectic pair*** on M is a tuple (ω, J), where ω is a non-degenerate 2-form on M, and J is ω-compatible, meaning that $\omega(\cdot, J\cdot)$ defines a J-invariant Riemannian metric, denoted by g_J (with ω implicit).

Definition 5.9. We say that a pair of almost symplectic pairs (ω_i, J_i), $i = 0, 1$, is δ***-close***, if ω_0, ω_1 are C^∞ δ-close, and J_0, J_1 are C^∞ δ-close.

Let $\mathcal{S}^k(A, \beta)$ denote the space of equivalence classes of all smooth, stable maps of nodal elliptic curves into M of charge class (A, β, k), with the topology defined by the quasi distance functions $\{\rho_\epsilon\}_\epsilon$ of McDuff–Salamon [28, 2nd ed., pages 143–144] on the set of stable maps, which involve the C^0 distance, the domains and the energy e_{g_J}, for g_J the metric of an almost symplectic pair (ω, J) on M as in Lemma 5.10 below; the genus 0 case is treated there, and the higher genus case is likewise well known. That is, elements of $\mathcal{S}^k(A, \beta)$ are like elements of $\overline{\mathcal{M}}_1^k(J, A, \beta)$ but are not required to be J-holomorphic, stable meaning that every component on which the map is constant is a stable marked curve. We use only the following three properties. (a) It has a basis $\mathcal{B}$ such that e_{g_J} is bounded on $\overline{V}$ for every $V \in \mathcal{B}$, and such that for every open W and $u \in W$ there is $V \in \mathcal{B}$ with $u \in V$ and $\overline{V} \subset W$. (b) Gromov convergence of J_j-holomorphic stable maps, $J_j \to J'$ in C^∞, for any almost complex structure J' on M, as in Theorem 5.2, is convergence in it. (c) For any smooth family $\{J_t\}$, $t \in [0, 1]$, of almost complex structures on M, the subspace of $\mathcal{S}^k(A, \beta) \times [0, 1]$ of pairs (u, t) with u J_t-holomorphic is metrizable, so that its subsets are compact if and only if they are sequentially compact; the subspace topology being the Gromov topology, see below, for a constant family in genus 0 this is the metrizability of the Gromov topology, [28, 2nd ed., Theorem 5.6.6], and the general case is likewise well known. We do not use metrizability of $\mathcal{S}^k(A, \beta)$ itself. By (a), (b), (c) and Theorem 5.2, for any almost complex structure J' on M the subspace topology on $\overline{\mathcal{M}}_1^k(J', A, \beta) \subset \mathcal{S}^k(A, \beta)$ is the Gromov topology: a sequence converging in $\mathcal{S}^k(A, \beta)$ has bounded energy by (a), hence a Gromov convergent subsequence, whose limit is the same by (b) and (c); the converse is (b); and the Gromov topology is first countable, its neighborhood bases in [32, Section 9.1] being indexed by a Hausdorff distance, so the two topologies, having the same convergent sequences, coincide. We denote the energy function by

$$e = e_{g_J} : \mathcal{S}^k(A, \beta) \to \mathbb{R}_{\geq 0}.$$

Lemma 5.10. *Let (ω, J) be an almost symplectic pair on a compact manifold M, α a closed integral 1-form on M, and let $N \subset \overline{\mathcal{M}}_1^k(J, A, \beta)$ be compact and open (as a subset of $\overline{\mathcal{M}}_1^k(J, A, \beta)$). Then there exists an open $U \subset \mathcal{S}^k(A, \beta)$ satisfying:*

(1) *e is bounded on $\overline{U}$.*

(2) *$U \supset N$.*

(3) $\overline{U} \cap \overline{\mathcal{M}}_1^k(J, A, \beta) = N$.

Proof. Let $\mathcal{B}$ be a basis of the topology of $\mathcal{S}^k(A, \beta)$ as in property (a) above.

Since N is relatively open, using the properties of $\mathcal{B}$ above, we may find a collection $\{V_a\} \subset \mathcal{B}$ s.t.

- $\{V_a\}$ covers N.
- $\overline{V}_a \cap \overline{\mathcal{M}}_1^k(J, A, \beta) \subset N$.

As N is compact, we have a finite subcover $\{V_{a_1}, \ldots, V_{a_n}\}$. Set $U := \cup_{i \in \{1,\ldots,n\}} V_{a_i}$. Then U satisfies the conclusion of the lemma. □

Lemma 5.11. *Let (M, ω, J, α) be as above, $N \subset \overline{\mathcal{M}}_1^k(J, A, \beta)$ an open compact set, and U as in the lemma above. Then there is a $\delta > 0$ s.t. whenever J' is C^∞ δ-close to J, if $u \in \overline{\mathcal{M}}_1^k(J', A, \beta)$ and $u \in \overline{U}$ then $u \in U$.*

Proof. Suppose otherwise, then there is a sequence $\{J_j\}$ C^∞ converging to J, and a sequence $\{u_j\} \subset \overline{U} \setminus U$ of J_j-holomorphic stable maps, $u_j \in \overline{\mathcal{M}}_1^k(J_j, A, \beta)$. Then by property 1 e_{g_J} is bounded on $\{u_j\}$. Hence, by Gromov compactness for the sequence $J_j \to J$, Theorem 5.2, we may find a Gromov convergent subsequence $\{u_{j_l}\}$ to a J-holomorphic stable map $u \in \overline{U} \setminus U$, by property (b), this set being closed, which has charge class (A, β, k) by Lemma 4.9, so that $u \in \overline{\mathcal{M}}_1^k(J, A, \beta)$. But by Properties 3, 2 of the set U,

$$(\overline{U} \setminus U) \cap \overline{\mathcal{M}}_1^k(J, A, \beta) = \emptyset.$$

So that we obtain a contradiction. □

Lemma 5.12. *Let M, ω, J, α and $N \subset \overline{\mathcal{M}}_1^k(J, A, \beta)$ be as in the previous lemma. Then there is a $\delta > 0$ s.t. the following is satisfied. Let (ω', J') be an almost symplectic pair δ-close to (ω, J), then there is a smooth family $\{J_t\}$, $J_0 = J$, $J_1 = J'$ s.t. there is an open compact subset*

$$\widetilde{N} \subset \overline{\mathcal{M}}_1^k(\{J_t\}, A, \beta),$$

satisfying

$$\widetilde{N} \cap (\overline{\mathcal{M}}_1^k(J, A, \beta) \times \{0\}) = N \times \{0\}.$$

Proof. Fix U as in Lemma 5.10, and let δ be as in Lemma 5.11 for this U. We then need:

Lemma 5.13. *Given a $\delta > 0$ there is a $\delta' > 0$ s.t. if (ω', J') is δ'-close to (ω, J) then there is a smooth family $\{(\omega_t, J_t)\}$, $t \in [0, 1]$, satisfying:*

- *(ω_t, J_t) is δ-close to (ω, J) for each t.*
- *$(\omega_0, J_0) = (\omega, J)$ and $(\omega_1, J_1) = (\omega', J')$.*

Proof. Let $\{g_t\}$ be the family of metrics on M given by the convex linear combination of $g = g_J, g' = g_{J'}$, $g_t = (1-t)g + tg'$. Likewise, the family of 2-forms $\{\omega_t\}$ given by the convex linear combination of ω, ω' is non-degenerate for each t if δ' was chosen to be sufficiently small. And each ω_t is C^∞ δ'-close to $\omega_0 = \omega$. We take $\delta' \leq \delta$.

Let

$$\mathrm{ret} : \mathrm{Met}(M) \times \Omega^{nd}(M) \to \mathcal{J}(M)$$

be the retraction map of [27, Proposition 2.50], where $\mathrm{Met}(M)$ is the space of metrics on M, $\Omega^{nd}(M)$ the space of non-degenerate 2-forms on M, and $\mathcal{J}(M)$ the space of almost complex structures. This map has the property that the almost complex structure $\mathrm{ret}(g, \omega)$ is compatible with ω, and that

$\mathrm{ret}(g_J, \omega) = J$ for $g_J = \omega(\cdot, J\cdot)$. Then $\{(\omega_t, \mathrm{ret}(g_t, \omega_t))\}$ is a compatible family. As ret is smooth, the family is smooth in t, and δ' can be chosen such that $\{\mathrm{ret}(g_t, \omega_t)\}$ are C^∞ δ-close to J. □

Returning to the proof of the main lemma. Let δ' be chosen as in Lemma 5.13 and let $\{(\omega_t, J_t)\}$ be the corresponding family. Set

$$\widetilde{N} = \overline{\mathcal{M}}_1^k(\{J_t\}, A, \beta) \cap (U \times [0,1]),$$

for U as fixed above.

Then $\widetilde{N}$ is an open subset of $\overline{\mathcal{M}}_1^k(\{J_t\}, A, \beta)$. By Lemma 5.11,

$$\widetilde{N} = \overline{\mathcal{M}}_1^k(\{J_t\}, A, \beta) \cap (\overline{U} \times [0,1]),$$

so that $\widetilde{N}$ is also closed.

Moreover $\sup_{(u,t)\in\widetilde{N}} e_{g_J}(u) < \infty$, by condition 1 of U, and the energies with respect to the metrics g_{ω_t, J_t} are then also bounded, these being uniformly comparable to g_J on the compact manifold M. Consequently $\widetilde{N}$ is sequentially compact: by Gromov compactness for sequences $J_{t_j} \to J_t$, Theorem 5.2, a sequence in $\widetilde{N}$ has a Gromov convergent subsequence, whose limit has charge class (A, β, k) by Lemma 4.9, is its limit in $\mathcal{S}^k(A, \beta) \times [0,1]$ by (b), and lies in $\widetilde{N}$, this being closed; hence it is compact by property (c) of the topology of $\mathcal{S}^k(A, \beta)$. Finally $\widetilde{N} \cap (\overline{\mathcal{M}}_1^k(J, A, \beta) \times \{0\}) = (\overline{U} \cap \overline{\mathcal{M}}_1^k(J, A, \beta)) \times \{0\} = N \times \{0\}$ by property 3, which gives the displayed equation. Resetting $\delta := \delta'$, we are then done with the proof of the main lemma. □

Proposition 5.14. *Let (ω, J) be an almost symplectic pair on a compact manifold M, α a closed integral 1-form on M, and suppose that $N \subset \overline{\mathcal{M}}_1^k(J, A, \beta)$ is open and compact, with $GW_1^k(N, J, A, \beta) \neq 0$. Then there is a $\delta > 0$ s.t. whenever (ω', J') is an almost symplectic pair δ-close to (ω, J), there exists $u \in \overline{\mathcal{M}}_1^k(J', A, \beta)$. In particular this holds whenever J' is C^∞ δ'-close to J, for a suitable $\delta' > 0$: the form $\omega'(v, w) := \frac{1}{2}(\omega(v, w) + \omega(J'v, J'w))$ is J'-invariant, satisfies $\omega'(v, J'v) = \omega(v, J'v) > 0$ for J' close to J, and is C^∞-close to ω.*

Proof. For N as in the hypothesis let δ be as in Lemma 5.12, and for (ω', J') δ-close to (ω, J) let $\{J_t\}$, $J_1 = J'$, and $\widetilde{N}$ be as given by that lemma. Then $N_0 = N$ for N_0, N_1 as in Lemma 5.6, and $c_1(A) = 0$ as $GW_1^k(N, J, A, \beta)$ is defined, so that lemma gives $GW_1^k(N_1, J', A, \beta) = GW_1^k(N, J, A, \beta) \neq 0$; hence $N_1 \neq \emptyset$. □

6. Elliptic curves in the lcs-fication of a contact manifold and the Fuller index

Throughout this section and the next, (C, λ) is a closed contact manifold. The following elementary result is crucial for us.

Lemma 6.1. *Let (λ, α, J) be a tamed first kind lcs pair on M. Then every non-constant closed (nodal) J-holomorphic curve $u : \Sigma \to M$ is a Reeb 2-curve.*

Proof of Lemma 6.1. Let $u : \Sigma \to M$ be a non-constant, nodal (possibly empty set of nodes) J-curve. By Lemma 4.5 the image of $d\widetilde{u}(z)$ lies in $\mathcal{V}_\lambda = \mathcal{D}$; being J-invariant, it is $\mathcal{D}$ wherever it is non-zero, $\mathcal{D}$ being 2-dimensional. So it is enough to show that $[u^*\alpha] \neq 0$. Let $\widetilde{M}$ denote the α-covering space of M, that is the space of equivalence classes of paths p starting at $x_0 \in M$, with a pair p_1, p_2 equivalent if $p_1(1) = p_2(1)$ and

$$\int_{[0,1]} p_1^*\alpha = \int_{[0,1]} p_2^*\alpha.$$

Then the lift of ω to $\widetilde{M}$ is

$$\widetilde{\omega} = \frac{1}{f} d(f\lambda),$$

where $f = e^{-g}$ and where g is a primitive for the lift $\widetilde{\alpha}$ of α to $\widetilde{M}$, that is $\widetilde{\alpha} = dg$. In particular $\widetilde{\omega}$ is conformally equivalent to an exact symplectic form on $\widetilde{M}$. So if $\widetilde{J}$ denotes the lift of J, any closed $\widetilde{J}$-curve is constant by Stokes theorem. Now if $[u^*\alpha] = 0$ then u has a lift to a $\widetilde{J}$-holomorphic map $v : \Sigma \to \widetilde{M}$. Since Σ is closed, it follows by the above that v is constant, so that u is constant, which is impossible. □

6.1. **Preliminaries on Reeb tori.** Let (λ, α) be the lcs-fication on $M = C \times S^1$ of the closed contact manifold (C, λ). For $\beta \in \pi_1(C)$ we set $A^1_\beta = \beta \otimes [S^1] \in H_2(M, \mathbb{Z})$, abbreviated A_β, for β also denoting the Hurewicz image of the class in $H_1(C, \mathbb{Z})$. Let $\mathcal{O}(R^\lambda, \beta)$ be the orbit space as in the Introduction. Let J^λ on $C \times S^1$ be as in Example 4.6 and Section 4.2. We have a map:

$$\mathcal{P} : \mathcal{O}(R^\lambda, \beta) \to \overline{\mathcal{M}}^1_1(J^\lambda, A^1_\beta, \beta), \quad \mathcal{P}(o) = u_o, \tag{6.2}$$

for u_o the Reeb torus as previously.

We set up some terminology. Let $pr_C : C \times S^1 \to C$ and $pr_{S^1} : C \times S^1 \to S^1$ be the projections. For $o \in \mathcal{O}(R^\lambda, \beta)$ let $m = m(o)$ be its multiplicity, Appendix A, so that $o = m\, o_s$ for a simple orbit o_s, where $d\, o$ denotes the d-fold iterate of an orbit string o, defined there, and set

$$\mathcal{T}_o := \text{image}\, u_o = \text{image}\, o_s \times S^1,$$

an embedded torus, tangent to the distribution $\mathcal{D} = \text{span}(R^\lambda, \partial_\theta)$. For $0 \le \ell < m$ define the ***sheared Reeb tori***

$$u_{o,\ell} : T^2 \to M, \quad u_{o,\ell}(s,t) = (o_s(ms + \ell t), t), \tag{6.3}$$

covering maps of degree m onto $\mathcal{T}_o$, of charge 1 and class A^1_β, with $u_{o,0} = u_o$.

Let $u : \Sigma \to C \times S^1$ be a map from a closed oriented surface and t_0 a regular value of $pr_{S^1} \circ u$; then each component S_0 of $(pr_{S^1} \circ u)^{-1}(t_0)$ is an embedded circle in Σ, oriented so that (w, v) is a positive basis of $T\Sigma$ for w a positive tangent vector of S_0 and v any vector with $d(pr_{S^1} \circ u)(v) > 0$; this is the convention $[\gamma] \cdot [\rho] = 1$ of Definition 4.7. We say that u ***traces*** o on S_0 if $pr_C \circ u|_{S_0}$ is, up to orientation preserving reparametrization of S_0, a representative of the string o.

The sheared Reeb torus $u_{o,\ell}$ traces o on every component of every fiber of $pr_{S^1} \circ u_{o,\ell}$, these fibers being the circles $S^1 \times \{t_0\}$. We can say more:

Proposition 6.4. *Let J be (λ, α)-admissible. Each $u_{o,\ell}$ is J-holomorphic for a unique complex structure on T^2, and every $u \in \overline{\mathcal{M}}^1_1(J, A^1_\beta, \beta)$ is the class of $u_{o,\ell}$ for a unique pair (o, ℓ), $o \in \mathcal{O}(R^\lambda, \beta)$, $0 \le \ell < m(o)$. Here o is the orbit string traced by u, in the sense above, on any component of any regular fiber of $pr_{S^1} \circ u$. In particular the map (6.2), with J in place of J^λ, is injective, with image*

$$\overline{\mathcal{M}}^{Reeb}(J, A^1_\beta, \beta) := \mathcal{P}(\mathcal{O}(R^\lambda, \beta)),$$

whose elements, the classes of the maps u_o, we also call **Reeb tori**. *(There is an analogous statement for $\overline{\mathcal{M}}^k_1(J, A^k_\beta, \beta)$, $k > 1$, where $A^k_\beta = k \cdot \beta \otimes [S^1]$.)*

In the particular case of J^λ, the domains of the Reeb tori are "rectangular", that is quotients of the complex plane by rectangular lattices, while the domains of the sheared Reeb tori (6.3), $\ell \neq 0$, need not be: the lattice of $u_{o,\ell}$ is $\mathbb{Z} + \mathbb{Z}(\frac{\ell}{m} + \frac{i}{c})$, for c the period of o. Rectangularity can also fail for the generalized Reeb tori of Section 7.1, for the mapping torus of some strict contactomorphism. Moreover, for non-admissible compatible almost complex structures the elements of $\overline{\mathcal{M}}^1_1(J, A_\beta, \beta)$ might be nodal.

Proof of Proposition 6.4. Since $\mathcal{T}_o$ is tangent to the J-invariant distribution $\mathcal{D}$, it is a J-holomorphic torus, so each covering map $u_{o,\ell} : T^2 \to \mathcal{T}_o$ is J-holomorphic for the unique pulled back complex structure; in particular $\mathcal{P}(o)$, the class of $u_o = u_{o,0}$, lies in $\overline{\mathcal{M}}_1^1(J, A^1_\beta, \beta)$. We now classify the elements of the moduli space. Suppose we have a curve $u \in \overline{\mathcal{M}}_1^1(J, A^1_\beta, \beta)$, represented by $u : \Sigma \to M$. By Lemma 6.1 u is a Reeb 2-curve. Then $\widetilde{\Sigma}$ has no spherical components: a non-constant one gives a J-holomorphic map $u' : \mathbb{CP}^1 \to M$, which by Lemma 6.1 is a Reeb 2-curve, impossible by the second property in Definition 1.12 as $H^1(\mathbb{CP}^1, \mathbb{R}) = 0$; and constant ones are excluded by stability, since they would form trees attached at single nodes to the genus one component of $\widetilde{\Sigma}$, which must then exist as u is non-constant, and the leaves of such a tree carry a single special point.

We now show that u is a covering map onto $\mathcal{T}_o$ for some o. As $\widetilde{\Sigma}$ has no spherical components, Σ is a smooth torus by the description of nodal elliptic curves in Section 4.1; that is, u is normal.

By the charge 1 condition $pr_{S^1} \circ u$ is surjective. By the Sard theorem we have a regular value $t_0 \in S^1$, so that $(pr_{S^1} \circ u)^{-1}(t_0)$ contains an embedded circle $S_0 \subset \Sigma$. Now $d(pr_{S^1} \circ u)$ is surjective onto $T_{t_0}S^1$ along $T\Sigma|_{S_0}$. And so by the first property of u being a Reeb 2-curve, $o = pr_C \circ u|_{S_0}$ has non-vanishing differential $d(o)$. Moreover, again by the first property, o is tangent to $\ker d\lambda$. It follows that o is an unparametrized λ-Reeb orbit. It is positively oriented. Let w be a positive tangent vector of S_0 and v as in the orientation convention, so that $jw = \epsilon v + \sigma w$ with $\epsilon > 0$. Then $du(w) = (aR^\lambda, 0)$ with $a \neq 0$, and $J(R^\lambda, 0) = (a'R^\lambda, b\,\partial_\theta)$ with $b = \omega((R^\lambda, 0), J(R^\lambda, 0)) > 0$, as J is admissible and $\omega((R^\lambda, 0), (0, \partial_\theta)) = \lambda(R^\lambda) = 1$. So $d(pr_{S^1} \circ u)(jw) = d(pr_{S^1})(Jdu(w)) = ab$, while $d(pr_{S^1} \circ u)(jw) = \epsilon\, d(pr_{S^1} \circ u)(v) > 0$. Hence $a > 0$.

Next, du vanishes at only finitely many points, the zeros of the differential of a non-constant J-holomorphic map being isolated, as for holomorphic maps, by the Carleman similarity principle [28, 2nd ed., Theorem 2.3.5, Lemma 2.4.1]; elsewhere $du(T\Sigma) = \mathcal{D}$. The distribution $\mathcal{D} = \operatorname{span}(R^\lambda, \partial_\theta)$ is integrable, with leaves $\gamma \times S^1$ for γ a Reeb trajectory. A connected immersed surface tangent to an integrable distribution lies in a single leaf, so $\operatorname{image} u \subset \gamma \times S^1$ for the trajectory γ through a point of o, that is $\operatorname{image} u \subset \mathcal{T}_o$.

As $\mathcal{T}_o$ is a J-holomorphic torus, $u : T^2 \to \mathcal{T}_o$ is a holomorphic map, hence a covering map, unbranched by the Riemann–Hurwitz formula as both tori have Euler characteristic 0. Coverings $T^2 \to \mathcal{T}_o$ are classified by the finite index subgroups of $\pi_1(\mathcal{T}_o) = \mathbb{Z}^2$, with generators the simple orbit o_s underlying o and the S^1 factor. The covering has charge one iff its subgroup surjects onto the second factor; its index is then the multiplicity m of the orbit traced by S_0, so $o = m\, o_s$. The index m subgroups surjecting onto the second factor are $\Lambda_\ell = \langle (m, 0), (\ell, 1) \rangle$, $0 \leq \ell < m$, and Λ_ℓ is the subgroup of the map $u_{o,\ell}$ of (6.3). Two coverings with the same subgroup are isomorphic as coverings, and the isomorphism is a biholomorphism for the pulled back complex structures. This proves the classification. Injectivity of $\mathcal{P}$ is automatic, and $\mathcal{P}(o) = u_{o,0}$, so $\mathcal{P}$ is a bijection onto the set of Reeb tori. □

Lemma 6.5. *For J (λ, α)-admissible, $\overline{\mathcal{M}}^{Reeb}(J, A^1_\beta, \beta)$ is open and closed in $\overline{\mathcal{M}}_1^1(J, A^1_\beta, \beta)$. The same holds for the subset of Reeb tori of a cobordism moduli space $\overline{\mathcal{M}}_1^1(\{J_t\}, A^1_\beta, \beta)$, for $\{J_t\}$ a smooth family of almost complex structures with each J_t admissible for the lcs-fication (λ_t, α) of a contact form λ_t on C, $\{\lambda_t\}$ a smooth family.*

Proof. By the proof of Proposition 6.4 (the subgroups Λ_ℓ) a curve u tracing o is a Reeb torus iff the subgroup $u_*\pi_1(T^2) \subset \pi_1(\mathcal{T}_o)$ contains the class of the S^1 factor. Suppose $u_i \to u$ in the Gromov topology, tracing o_i and o; then $o_i \to o$ in $\mathcal{O}(R^\lambda, \beta)$, as the fibers of $pr_{S^1} \circ u_i$ converge to those of $pr_{S^1} \circ u$. For i large the nearest point projection $\pi_i : \mathcal{T}_{o_i} \to \mathcal{T}_o$ is a covering map taking the class of the S^1 factor to the class of the S^1 factor. The maps $\pi_i \circ u_i$ are C^0 close to u after reparametrization, hence homotopic to u as maps into $\mathcal{T}_o$. So $u_*\pi_1(T^2) = (\pi_i \circ u_i)_*\pi_1(T^2)$, and this contains the S^1 class iff $u_{i*}\pi_1(T^2)$ does, as π_{i*} is injective and preserves the second coordinate. Thus the set of Reeb tori

and its complement are both closed. The cobordism case is identical, with $\mathcal{T}_{o_i}$ the images of the curves at times $t_i \to t$. □

For J (λ,α)-admissible, as in Proposition 6.4, we also have the map

$$\mathrm{Orb} : \overline{\mathcal{M}}_1^1(J, A_\beta^1, \beta) \to \mathcal{O}(R^\lambda, \beta), \quad u_{o,\ell} \mapsto o, \tag{6.6}$$

a left inverse of $\mathcal{P}$, whose fiber over o consists of the $m(o)$ curves (6.3). It is continuous, by the argument of Lemma 6.5, and proper: the curves over a compact set of orbit strings have energy bounded by the periods, so a sequence of them has a Gromov convergent subsequence by Theorem 5.2, whose limit lies in $\overline{\mathcal{M}}_1^1(J, A_\beta^1, \beta)$ by Lemma 4.9; cf. Proposition 7.1. Thus for an open compact $N \subset \mathcal{O}(R^\lambda, \beta)$ both $\mathrm{Orb}^{-1}(N)$ and $\mathcal{P}(N) = \mathrm{Orb}^{-1}(N) \cap \overline{\mathcal{M}}^{Reeb}(J, A_\beta^1, \beta)$ are open compact subsets of $\overline{\mathcal{M}}_1^1(J, A_\beta^1, \beta)$, the latter by Lemma 6.5.

Proposition 6.7. *Let (C,λ) be a closed contact manifold and o a period c closed R^λ-Reeb orbit. The kernel of the normal Cauchy–Riemann operator $D : \Omega^0(\mathcal{W}) \to \Omega^{0,1}(\mathcal{W})$ of the Reeb torus u_o, defined in the proof, is naturally identified with the 1-eigenspace of $F^\lambda_{c,*}$, the time c linearized return map $\xi(o(0)) \to \xi(o(0))$ induced by the R^λ Reeb flow. In particular, if o is non-degenerate then u_o is a regular curve, that is the operator (6.8) below is surjective at it; so if λ is non-degenerate, meaning that all its closed Reeb orbits are, all the elements of $\overline{\mathcal{M}}^{Reeb}(J^\lambda, A_\beta, \beta)$ are regular.*

Proof. By definition every $u \in \overline{\mathcal{M}}^{Reeb}(J^\lambda, A_\beta, \beta)$ is a Reeb torus. In particular, such curves have a representation by a J^λ-holomorphic map

$$u : (T^2, j) \to (Y = C \times S^1, J^\lambda).$$

Since each u is immersed we may naturally get a splitting $u^*T(Y) \simeq \mathcal{W} \oplus T(T^2)$, using the metric $\omega(\cdot, J^\lambda \cdot)$, where $\mathcal{W} \to T^2$ denotes the pull-back of the normal bundle to image u for this metric, and which is identified with the pullback of the co-vanishing distribution ξ_λ on Y.

The full associated real linear Cauchy–Riemann operator takes the form:

$$D_u^J : \Omega^0(\mathcal{W} \oplus T(T^2)) \oplus T_j\mathcal{S} \to \Omega^{0,1}(\mathcal{W} \oplus T(T^2)). \tag{6.8}$$

Here $\mathcal{S} \subset \mathcal{J}(T^2)$ is a Teichmüller slice through j, [46, Definition 3.2]: a 2-dimensional family of complex structures through j representing the Teichmüller space of the torus near j, which enters (6.8) through $y \mapsto \frac{1}{2} J^\lambda\, du \circ y$ for $y \in T_j\mathcal{S}$. This is an index 2 Fredholm operator (after standard Sobolev completions), whose restriction to $\Omega^0(\mathcal{W} \oplus T(T^2))$ preserves the splitting, that is the restricted operator splits as

$$D \oplus D' : \Omega^0(\mathcal{W}) \oplus \Omega^0(T(T^2)) \to \Omega^{0,1}(\mathcal{W}) \oplus \Omega^{0,1}(T(T^2)).$$

The splitting holds since $\omega(\cdot, J^\lambda \cdot)$ is the product of the metric $\lambda^2 + d\lambda(\cdot, J\cdot)$ on C with $d\theta^2$, for which R^λ is a unit geodesic field with $\nabla_\nu R^\lambda \in \xi$ for $\nu \in \xi$, so that the linearized operator takes sections of $\mathcal{W}$ to $\Omega^{0,1}(\mathcal{W})$, while sections of $T(T^2)$ go to $\Omega^{0,1}(T(T^2))$ for any immersed J-curve, as $u \circ \phi_\tau$ is J-holomorphic for $\phi_\tau^* j$, for ϕ_τ the flow of a vector field on T^2. Only this splitting is used below. On the other hand the restricted Fredholm index 2 operator

$$\Omega^0(T(T^2)) \oplus T_j\mathcal{S} \to \Omega^{0,1}(T(T^2)),$$

is surjective by classical Teichmüller theory, see also [46, Lemma 3.1, Definition 3.2] for a precise argument in this setting. It follows that D_u^J will be surjective if the restricted Fredholm index 0 operator

$$D : \Omega^0(\mathcal{W}) \to \Omega^{0,1}(\mathcal{W}),$$

has no kernel.

The bundle $\mathcal{W}$ is symplectic with symplectic form on the fibers given by restriction of $u^* d\lambda$, and together with J^λ this gives a Hermitian structure (g_λ, j_λ) on $\mathcal{W}$. We have a linear symplectic connection $\mathcal{A}$ on $\mathcal{W}$, which over the slices $S^1 \times \{t\} \subset T^2$ is induced by the pullback by u of the linearized R^λ

Reeb flow. Specifically the $\mathcal{A}$-transport map from the fiber $\mathcal{W}_{(s_0,t)}$ to the fiber $\mathcal{W}_{(s_1,t)}$ over the path $[s_0,s_1]\times\{t\}\subset T^2$, is given by

$$(u_*|_{\mathcal{W}_{(s_1,t)}})^{-1}\circ(F^\lambda_{c(s_1-s_0)})_*\circ u_*|_{\mathcal{W}_{(s_0,t)}},$$

where $F^\lambda_{c(s_1-s_0)}$ is the time $c\cdot(s_1-s_0)$ map for the R^λ Reeb flow, where c is the period of the Reeb orbit o with $u=u_o$, and where $u_*:\mathcal{W}\to TY$ denotes the natural map (it is the universal map in the pull-back diagram).

The connection $\mathcal{A}$ is defined to be trivial in the t direction, where trivial means that the parallel transport maps along the segments $\{s\}\times[t_0,t_1]$ are the identity under the identifications $\mathcal{W}_{(s,t)}=\xi(o(s))$. In particular the curvature $R_{\mathcal{A}}$, understood as a Lie algebra valued 2-form, of this connection vanishes. The connection $\mathcal{A}$ determines a real linear CR operator $D_{\mathcal{A}}$ on $\mathcal{W}$ in the standard way, take the complex anti-linear part of the vertical differential of a section. Explicitly,

$$D_{\mathcal{A}}:\Omega^0(\mathcal{W})\to\Omega^{0,1}(\mathcal{W}),$$

is defined by

$$D_{\mathcal{A}}(\mu)(p)=\tfrac{1}{2}\left(\pi^{vert}(\mu(p))\circ d\mu(p)+j_\lambda\circ\pi^{vert}(\mu(p))\circ d\mu(p)\circ j\right),$$

where

$$\pi^{vert}(\mu(p)):T_{\mu(p)}\mathcal{W}\to T^{vert}_{\mu(p)}\mathcal{W}\simeq\mathcal{W}$$

is the $\mathcal{A}$-projection, and where $T^{vert}_{\mu(p)}\mathcal{W}$ is the kernel of the projection $T_{\mu(p)}\mathcal{W}\to T_pT^2$.

In fact $D=D_{\mathcal{A}}$. The operators D and $D_{\mathcal{A}}$ act between the same spaces. They have the same principal symbol, namely $\overline{\partial}$ for the Hermitian structure (g_λ,j_λ) on $\mathcal{W}$. So their difference is C^∞-linear in the section. That is, it is a zeroth order operator, given by a bundle map $\mathcal{W}\to\Lambda^{0,1}\otimes\mathcal{W}$. A zeroth order operator is compact after the Sobolev completions. Hence D and $D_{\mathcal{A}}$ are Fredholm of the same index. That index is 0, since $\mathcal{W}$ is pulled back from the circle $S^1\times\{0\}$, so that $c_1(\mathcal{W})=0$. Finally, the bundle map vanishes, so that $D=D_{\mathcal{A}}$. Indeed, for the Levi-Civita connection ∇ of the product metric above, $D\zeta=\frac{1}{2}(\nabla\zeta+J\nabla\zeta\circ j)-\frac{1}{2}J(\nabla_\zeta J)\,du$ for $\zeta\in\Omega^0(\mathcal{W})$, [28, 2nd ed., Proposition 3.1.1]. Here $\nabla_{\partial_s}\zeta=c\nabla_{R^\lambda}\zeta=c(L_{R^\lambda}\zeta+\nabla_\zeta R^\lambda)$, as ∇ is torsion free, $\nabla_{\partial_t}\zeta=\partial_\theta\zeta$, and $(\nabla_\zeta J)R^\lambda=-J\nabla_\zeta R^\lambda$, as $JR^\lambda=\partial_\theta$ is parallel. So $D\zeta(\partial_s)=\frac{c}{2}(L_{R^\lambda}\zeta+J\partial_\theta\zeta)=D_{\mathcal{A}}\zeta(\partial_s)$, the $\mathcal{A}$-transport along ∂_s being the linearized Reeb flow.

This is the standard computation of the linearized operator at a trivial cylinder over a Reeb orbit in a symplectization, of which u_o is the $\mathbb{Z}$-quotient: $(C\times S^1,J^\lambda)$ is the quotient of $\mathbb{R}\times C$, with $\theta=-r$ as in Section 1.1 and the cylindrical almost complex structure $J\partial_r=R^\lambda$, $J=J^\lambda$ on ξ, by $r\mapsto r+1$; there the linearized operator at the trivial cylinder $(s,t)\mapsto(cs,o(t))$, holomorphic for $j\partial_s=\partial_t$ as $J\partial_r=R^\lambda$, is $\partial_s+J\nabla_t$, with t the orbit parameter, that is with the roles of s and t exchanged relative to u_o, for ∇ the connection on ξ along o whose parallel transport is the linearized Reeb flow, see Wendl [44, Definition 3.22, Remark 3.24]. See also [30, Section 10.1] for a computation of this kind in much greater generality.

We have a differential 2-form Ω on the total space of $\mathcal{W}$ defined as follows. On the fibers $T^{vert}\mathcal{W}$, $\Omega=(u_*)^*\omega$, for $\omega=d_\alpha\lambda$, and for $T^{vert}\mathcal{W}\subset T\mathcal{W}$ denoting the vertical tangent space, or subspace of vectors v with $\pi_*v=0$, for $\pi:\mathcal{W}\to T^2$ the projection. While on the $\mathcal{A}$-horizontal distribution Ω is defined to vanish, and $\Omega(v,h):=0$ for v vertical and h horizontal. The 2-form Ω is closed, which we may check explicitly by using that $R_{\mathcal{A}}$ vanishes to obtain local symplectic trivializations of $\mathcal{W}$ in which $\mathcal{A}$ is trivial. Clearly $\int_0\Omega=0$ over the 0-section, since it is an $\mathcal{A}$-flat section. But any section is homotopic to the 0-section and Ω is closed, so in particular if $\mu\in\ker D$ then $\int_\mu\Omega=0$.

Since $\mu\in\ker D$, and so its vertical differential is complex linear, it follows that the vertical differential vanishes. To see this note that $\Omega(v,j_\lambda v)>0$, for $0\neq v\in T^{vert}\mathcal{W}$ and so if the vertical differential did not vanish we would have $\int_\mu\Omega>0$. So μ is $\mathcal{A}$-flat, in particular the restriction of μ over all slices $S^1\times\{t\}$ is identified with a period c orbit of the R^λ Reeb flow linearized at o, and which does not

depend on t as $\mathcal{A}$ is trivial in the t variable. So the kernel of D is identified with the vector space of period c orbits of the R^λ Reeb flow linearized at o, as needed. □

Proposition 6.9. *Let λ be a contact form on a $(2n+1)$-fold C, and o a non-degenerate, period c, λ-Reeb orbit, then the orientation of $[u_o]$ induced by the determinant line bundle orientation of $\overline{\mathcal{M}}_1^1(J^\lambda, A_\beta, \beta)$, is $(-1)^{CZ(o)-n}$, which is*

$$\operatorname{sign}\operatorname{Det}(\operatorname{id}|_{\xi(o(0))} - F^\lambda_{c,*}|_{\xi(o(0))}).$$

The sign of a Reeb torus is thus the fixed point index of the return map of its orbit. This is the analogue of the computation of Ionel–Parker [20, Lemma 4.6] for the symplectic mapping torus of a symplectomorphism, where the torus over a fixed point counts with its Lefschetz sign; there it is obtained by gluing, here by spectral flow.

Proof of Proposition 6.9. Abbreviate u_o by u. Let $\mathcal{W} \to T^2$ be the vector bundle associated to u as in the proof of Proposition 6.7. Fix a unitary trivialization τ of $\mathcal{W}$ induced by a unitary trivialization of the contact distribution $(\xi, d\lambda, J)$ along o in the obvious sense: $\mathcal{W}$ is the pullback of ξ along the composition

$$T^2 \to S^1 \xrightarrow{o} C.$$

Let the symplectic connection $\mathcal{A}$ on $\mathcal{W}$ be defined as before. Then the push-forward connection $\mathcal{A}' := \tau_*\mathcal{A} = (\tau^{-1})^*\mathcal{A}$ on $T^2 \times \mathbb{R}^{2n}$ is a connection whose parallel transport paths $p : [0,1] \to \operatorname{Symp}(\mathbb{R}^{2n})$, along the closed loops $S^1 \times \{t\}$, start at id and are independent of t. The parallel transport of $\mathcal{A}'$ along $\{s\} \times S^1$ is the identity, as $\mathcal{A}$ is trivial in the t direction by definition and τ is pulled back from the circle, that is $\mathcal{A}'$ is trivial in the t variable. We shall call such a connection $\mathcal{A}'$ on $T^2 \times \mathbb{R}^{2n}$ *induced by* p.

By non-degeneracy assumption on o, the map $p(1)$ has no 1-eigenvalues. Let $p'' : [0,1] \to \operatorname{Symp}(\mathbb{R}^{2n})$ be a path from $p(1)$ to a unitary map $p''(1)$, with $p''(1)$ having no 1-eigenvalues, and s.t. p'' has only simple and regular crossings with the Maslov cycle, that is at each crossing time the 1-eigenspace is one-dimensional and the crossing form on it is non-zero. Let p' be the concatenation of p and p''. The path p' is homotopic relative end points to a unitary geodesic path h starting at id and having regular crossings, so $CZ(p') = CZ(h)$. At each interior crossing of h the crossing form has even signature, by unitarity. At time 0 the crossing form of h is $\Gamma(h,0) = \bigoplus_j \vartheta_j |z_j|^2$ on $\mathbb{C}^n$, for $h(t) = \bigoplus_j e^{i\vartheta_j t}$ with all $\vartheta_j \neq 0$, so $\frac{1}{2}\operatorname{sign}\Gamma(h,0)$ is the number of positive ϑ_j minus the number of negative ones, which is congruent to n modulo 2. Hence

$$CZ(p') \equiv CZ(h) \equiv n \mod 2,$$

in the notation of [33]. Consequently,

$$CZ(p'') \equiv CZ(p) - n \mod 2, \tag{6.10}$$

by additivity of the Conley–Zehnder index.

Let us then define a free homotopy $\{p_t\}$ of p to p', p_t is the concatenation of p with $p''|_{[0,t]}$, all paths being reparametrized to be constant near their endpoints, reparametrized to have domain $[0,1]$ at each moment t. This determines a homotopy $\{\mathcal{A}'_t\}$ of connections induced by $\{p_t\}$. By the proof of Proposition 6.7, the CR operator D_t determined by each $\mathcal{A}'_t$ is surjective except at some finite collection of times $t_i \in (0,1)$, $i \in \{1, \dots, \nu\}$ determined by the crossing times of p'' with the Maslov cycle, and the dimension of the kernel of D_{t_i} is the dimension of the 1-eigenspace of $p''(t_i)$, which is 1 by the assumption that the crossings of p'' are simple.

The operator D_1 need not be complex linear. To fix this we concatenate the homotopy $\{D_t\}$ with the homotopy $\{\widetilde{D}_t\}$ defined as follows. Let $\{\widetilde{\mathcal{A}}_t\}$ be a homotopy of $\mathcal{A}'_1$ to a unitary connection $\widetilde{\mathcal{A}}_1$, where the homotopy $\{\widetilde{\mathcal{A}}_t\}$ is through connections induced by paths $\{\widetilde{p}_t\}$, giving a path homotopy of $p' = \widetilde{p}_0$

to h; all paths are reparametrized to be constant near their endpoints, so that the induced connections are smooth. Then $\{\widetilde{D}_t\}$ is defined to be induced by $\{\widetilde{\mathcal{A}}_t\}$.

Let us denote by $\{D_t^\sharp\}$ the concatenation of $\{D_t\}$ with $\{\widetilde{D}_t\}$. By construction, in the second half of the homotopy $\{D_t^\sharp\}$, $D_t^\sharp$ is surjective. And $D_1^\sharp$ is induced by a unitary connection, since it is induced by the unitary path $\widetilde{p}_1$. Consequently, $D_1^\sharp$ is complex linear. By the above construction, for the homotopy $\{D_t^\sharp\}$, $D_t^\sharp$ is surjective except for ν times in $(0,1)$, where the kernel has dimension one. Each of these crossings is transversal, that is the orientation of the determinant line changes sign there: the kernel of $D_t^\sharp$ is the 1-eigenspace of the monodromy $p_t(1)$, and the crossing form of the spectral flow of $\{D_t^\sharp\}$ is that of the path of monodromies, which is non-zero by regularity, Robbin–Salamon [34]: restricted to the kernel, the flat sections, $\langle \mu, \dot{D}_t^\sharp \mu\rangle_{L^2} = \frac{1}{2}\int \omega_0(\mu, \dot{a}_t\mu)$ for the connection forms a_t, which is the crossing form of the monodromies. Also $CZ(p'') \equiv \nu \mod 2$, each simple regular crossing of p'' contributing ± 1, while $p''(0)$ and $p''(1)$ have no 1-eigenvalue. In particular the sign of $[u]$ by the definition via the determinant line bundle is exactly

$$(-1)^\nu = (-1)^{CZ(p)-n},$$

by (6.10), which is what was to be proved. □

The sheared Reeb tori (6.3) are handled by the same argument.

Lemma 6.11. *Let λ be a contact form on a $(2n+1)$-fold C, and let $o = m\,o_s$ be a λ-Reeb orbit of multiplicity m, with o_s simple of period c_s. Let $P := F^\lambda_{c_s,*}|_{\xi(o(0))}$ be the linearized return map of o_s. For the sheared Reeb torus $u_{o,\ell}$, $0 \le \ell < m$, set $d := \gcd(\ell, m)$, with $\gcd(0,m) = m$. Then the kernel of the real linear Cauchy–Riemann operator $D : \Omega^0(\mathcal{W}) \to \Omega^{0,1}(\mathcal{W})$ of $u_{o,\ell}$, defined as in the proof of Proposition 6.7, is naturally identified with $\ker(P^d - \mathrm{id})$, the 1-eigenspace of the linearized return map of $d\,o_s$. In particular, if o is non-degenerate then so is $d\,o_s$, as $\ker(P^d - \mathrm{id}) \subset \ker(P^m - \mathrm{id})$, $u_{o,\ell}$ is regular, and its orientation sign is*

$$(-1)^{CZ(d\,o_s)-n} = i(d\,o_s),$$

in the notation of (A.1).

Proof. We have $u_{o,\ell} = u_{o_s} \circ \kappa_{m,\ell}$, for $u_{o_s}(\sigma, t) = (o_s(\sigma), t)$ the Reeb torus of o_s and $\kappa_{m,\ell} : T^2 \to T^2$, $\kappa_{m,\ell}(s,t) = (ms + \ell t, t)$, a covering map of degree m, which is holomorphic for the complex structures on the two tori for which $u_{o,\ell}$ and u_{o_s} are J^λ-holomorphic, as u_{o_s} is an immersion. Hence $\mathcal{W} = \kappa_{m,\ell}^*\mathcal{W}_{o_s}$ and $D = \kappa_{m,\ell}^* D_{o_s}$, for $\mathcal{W}_{o_s}$, D_{o_s} the normal bundle and normal Cauchy–Riemann operator of u_{o_s}. Let $\mathcal{A}_{o_s}$ be the connection on $\mathcal{W}_{o_s}$ of the proof of Proposition 6.7: parallel transport by the linearized R^λ flow along the projection to C of paths in T^2. Its holonomy along the loop $S^1 \times \{t\}$ is P, and along $\{\sigma\} \times S^1$ it is the identity. Set $\mathcal{A} := \kappa_{m,\ell}^*\mathcal{A}_{o_s}$. The loops $s \mapsto (s,t)$ and $t \mapsto (s,t)$ in the domain of $u_{o,\ell}$ are mapped by $\kappa_{m,\ell}$ to loops in the classes $(m,0)$ and $(\ell,1)$ of $\pi_1(T^2) = \mathbb{Z}^2$, so the holonomies of $\mathcal{A}$ along them are P^m and P^ℓ. As $D_{o_s} = D_{\mathcal{A}_{o_s}}$ by the proof of Proposition 6.7, we get $D = \kappa_{m,\ell}^* D_{\mathcal{A}_{o_s}} = D_{\mathcal{A}}$, and the argument there, with the pulled back 2-form Ω, shows that every $\mu \in \ker D$ is $\mathcal{A}$-flat. So $\ker D$ is identified with the subspace of $\xi(o(0))$ fixed by both holonomies, which is $\ker(P^m - \mathrm{id}) \cap \ker(P^\ell - \mathrm{id}) = \ker(P^d - \mathrm{id})$, since d divides m and ℓ and is an integer combination of them.

For the sign we run the proof of Proposition 6.9 with p the path of the linearized R^λ flow along o_s over one period, so that $p(1) = P$, choosing the path p'' generic: $p''(1)$ has no eigenvalue which is a d-th root of unity, and the path $t \mapsto p''(t)^d$ has only regular crossings with the Maslov cycle. Their kernels need not be one-dimensional, as the eigenvalues $e^{\pm 2\pi i j/d}$ cross together; at a regular crossing with k-dimensional kernel the orientation of the determinant line changes by $(-1)^k$, for both operators below. Along the resulting homotopy of connections, the operator induced on $\kappa_{m,\ell}^*\mathcal{W}_{o_s}$ has kernel identified, by the above, with the 1-eigenspace of $p_t(1)^d$, with crossing forms m/d times those of the

operator induced on the normal bundle of the Reeb torus $u_{d\,o_s} = u_{o_s} \circ \kappa_{d,0}$ of the d-fold cover, by the pullbacks under $\kappa_{d,0}$ of the same connections, whose kernel is again the 1-eigenspace of $p_t(1)^d$. Indeed $\Lambda_\ell \subset \langle (d,0),(0,1)\rangle$, so $\kappa_{m,\ell} = \kappa_{d,0}\circ\kappa'$ for a covering κ' of degree m/d; the kernel elements on $\kappa_{m,\ell}^*\mathcal{W}_{o_s}$ are the pullbacks under κ' of those on $\kappa_{d,0}^*\mathcal{W}_{o_s}$, both being the flat sections, and L^2 pairings multiply by $\deg\kappa'$. Both homotopies end at complex linear operators. So the two spectral flows coincide, and the sign of $u_{o,\ell}$ is that of $u_{d\,o_s}$, which is $(-1)^{CZ(d\,o_s)-n}$ by Proposition 6.9. □

For an open compact $N \subset \mathcal{O}(R^\lambda,\beta)$ and $e \geq 1$ let

$$N^{(e)} := \{o' \in \mathcal{O}(R^\lambda,\gamma) \,|\, e\gamma = \beta,\ e\,o' \in N\}$$

be the set of e-th roots of elements of N, where $e\,o'$ is the e-fold iterate of o' and $e\gamma$ the class of the e-fold iterate of a loop in class γ. Iteration $o' \mapsto e\,o'$ is continuous and proper on the space of all orbit strings, so $N^{(e)}$ is open and compact and meets only finitely many classes γ; we write $i(N^{(e)},R^\lambda)$ for the sum of the Fuller indices of its intersections with these $\mathcal{O}(R^\lambda,\gamma)$. Clearly $N^{(1)} = N$, and $N^{(e)} = \emptyset$ once e exceeds the multiplicity of every element of N; the multiplicity is bounded on N, as the period is continuous, hence bounded on N, while the periods of simple orbits are bounded below. Set

$$\hat{i}(N,R^\lambda,\beta) := \sum_{e\geq 1} \frac{\varphi(e)}{e}\, i(N^{(e)},R^\lambda), \tag{6.12}$$

for φ Euler's totient function. When N is finite and non-degenerate, writing $o = m(o)\,o_s$ with o_s simple, and using that $o \mapsto o/e := (m(o)/e)\,o_s$ is a bijection from $\{o \in N \,|\, e \mid m(o)\}$ onto $N^{(e)}$ with $m(o/e) = m(o)/e$, this reads

$$\hat{i}(N,R^\lambda,\beta) = \sum_{o\in N} \frac{1}{m(o)} \sum_{d|m(o)} \varphi\left(\tfrac{m(o)}{d}\right) i(d\,o_s) = \sum_{o\in N} \frac{1}{m(o)} \sum_{g\in\mathbb{Z}/m(o)} i\left((m(o)/\operatorname{ord}(g))\,o_s\right), \tag{6.13}$$

the last sum being over the stabilizer $\mathbb{Z}/m(o) \subset S^1$ of o under reparametrization. So $\hat{i}$, which we call the ***inertia Fuller index*** of N, counts pairs (o,g), g in the stabilizer of o, each weighted by $1/m(o)$ times the fixed point index of the quotient orbit string $o/\langle g\rangle = (m(o)/\operatorname{ord}(g))\,o_s$: it is a Lefschetz number on the inertia groupoid of the orbit space. It coincides with $i(N,R^\lambda,\beta)$ when all elements of N are simple.

Theorem 6.14. *Let (C,λ) be a closed contact manifold, $N \subset \mathcal{O}(R^\lambda,\beta)$ open and compact, and let $\mathcal{P}$,* Orb *be as in (6.2), (6.6), for $J = J^\lambda$. Then*

$$GW_1^1(\mathcal{P}(N),J^\lambda,A_\beta,\beta) = i(N,R^\lambda,\beta), \tag{6.15}$$

$$GW_1^1(\mathrm{Orb}^{-1}(N),J^\lambda,A_\beta,\beta) = \hat{i}(N,R^\lambda,\beta), \tag{6.16}$$

where i is the Fuller index of Appendix A, $\hat{i}$ is as in (6.12), and the left-hand sides are the counts (5.5).

Proof. Suppose first that N is finite and consists of non-degenerate orbits $\{o_i\}$, with multiplicities $m(o_i)$ as in Appendix A, and write $o_i = m(o_i)\,o_{i,s}$ with $o_{i,s}$ simple. By Propositions 6.7 and 6.9 the Reeb tori u_{o_i} are regular with sign $(-1)^{CZ(o_i)-n}$, where $\dim C = 2n+1$ and the sign is as in (A.1), and by Lemma 6.11 the sheared Reeb tori $u_{o_i,\ell}$ are regular with sign $i(\gcd(\ell,m(o_i))\,o_{i,s})$. Each of these curves has isotropy group of order $m(o_i)$: the automorphisms of $u_{o_i,\ell} = u_{o_{i,s}} \circ \kappa_{m(o_i),\ell}$, in the notation of Lemma 6.11, are the deck transformations of the covering $\kappa_{m(o_i),\ell}$, which is normal of degree $m(o_i)$ as $\pi_1(T^2)$ is abelian; cf. Appendix B. Hence

$$GW_1^1(\mathcal{P}(N),J^\lambda,A_\beta,\beta) = \sum_i \frac{(-1)^{CZ(o_i)-n}}{m(o_i)} = i(N,R^\lambda,\beta),$$

and

$$GW_1^1(\mathrm{Orb}^{-1}(N),J^\lambda,A_\beta,\beta) = \sum_i \frac{1}{m(o_i)} \sum_{\ell\in\mathbb{Z}/m(o_i)} i\left(\gcd(\ell,m(o_i))\,o_{i,s}\right) = \hat{i}(N,R^\lambda,\beta),$$

by (6.13), as there are $\varphi(m/d)$ elements $\ell \in \mathbb{Z}/m$ with $\gcd(\ell, m) = d$. Thus, the theorem follows for N as above.

In general, by a C^1-small perturbation we obtain a smooth family $\{\lambda_t\}$ of contact forms, $\lambda_0 = \lambda$, with λ_1 non-degenerate, that is with non-degenerate orbits, which is possible as non-degenerate contact forms are C^∞-generic, see [16], whose argument is independent of the dimension, together with a smooth family $\{J^{\lambda_t}\}$ as in Example 4.6, obtained by choosing S^1-invariant $d\lambda_t$-compatible structures on $\ker \lambda_t$ smoothly in t, which is possible as the space of such structures is contractible, and, by [38, Lemma 4.1], an open compact subset $\widetilde{N}_{\mathcal{O}}$ of $S(\{R^{\lambda_t}\}, \beta)/S^1$, in the notation of Appendix A, consisting of the class β orbit strings of the R^{λ_t} near N, with $\widetilde{N}_{\mathcal{O}} \cap (L_\beta C \times \{0\})/S^1 = N$. Set $N_1 := \widetilde{N}_{\mathcal{O}} \cap (L_\beta C \times \{1\})/S^1$, a finite set of non-degenerate orbits. By the invariance of the Fuller index, $i(N_1, R^{\lambda_1}, \beta) = i(N, R^\lambda, \beta)$; likewise $i(N_1^{(e)}, R^{\lambda_1}) = i(N^{(e)}, R^\lambda)$ for every e, applying the invariance to the open compact set of e-th roots of elements of $\widetilde{N}_{\mathcal{O}}$, whose slices at 0 and 1 are $N^{(e)}$ and $N_1^{(e)}$; so also $\hat{i}(N_1, R^{\lambda_1}, \beta) = \hat{i}(N, R^\lambda, \beta)$. On the other hand, the maps Orb of (6.6) for the J^{λ_t} assemble to a continuous proper map $\widetilde{\mathrm{Orb}} : \overline{\mathcal{M}}_1^1(\{J^{\lambda_t}\}, A_\beta, \beta) \to S(\{R^{\lambda_t}\}, \beta)/S^1$, $(u, t) \mapsto (\mathrm{Orb}(u), t)$, as in the proof of Theorem 1.8. So $\widetilde{N} := \widetilde{\mathrm{Orb}}^{-1}(\widetilde{N}_{\mathcal{O}})$ is an open compact subset of $\overline{\mathcal{M}}_1^1(\{J^{\lambda_t}\}, A_\beta, \beta)$, with slices $\mathrm{Orb}^{-1}(N)$ at $t = 0$ and $\mathrm{Orb}^{-1}(N_1)$ at $t = 1$. Intersecting $\widetilde{N}$ with the open and closed subset of Reeb tori of Lemma 6.5, we get the same with $\mathcal{P}(N)$ and $\mathcal{P}(N_1)$ in place of $\mathrm{Orb}^{-1}(N)$ and $\mathrm{Orb}^{-1}(N_1)$. Then by Lemma 5.6 the left-hand sides of (6.15) and (6.16) do not change when (N, λ) is replaced by (N_1, λ_1), and the theorem follows from the finite non-degenerate case. □

What about higher genus invariants of $C \times S^1$? Following the proof of Proposition 6.4, it is not hard to see that all J^λ-holomorphic curves must be branched covers of sheared Reeb tori. If one can show that these branched covers are regular when the underlying tori are regular, the calculation of invariants would be fairly automatic from this data. See [47], [45] where these kinds of regularity calculation are made.

7. Proofs of main theorems

7.1. **Mapping tori and Reeb 2-curves.** Let (C, λ) be a closed contact manifold and ϕ a strict contactomorphism and let $M = (M_{\phi,1}, \lambda_\phi, \alpha)$ denote the mapping torus of ϕ, as also appearing in Theorem 2.10. More specifically, $M = C \times \mathbb{R}/\sim$, where the equivalence $\sim$ is generated by $(x, \theta) \sim (\phi(x), \theta + 1)$, with λ_ϕ and $\alpha = d\theta$ as in Example 2.9 for $c = 1$; see also [3]. Then (λ_ϕ, α) is an integral first kind lcs pair on M.

In this case $\mathcal{V}_{\lambda_\phi} = \mathcal{D}$, and in mapping torus coordinates at a point (x, θ), it is spanned by $X_{\lambda_\phi} = (0, -\frac{\partial}{\partial\theta}), X_\alpha = (R^{\lambda_\theta}, 0)$ for R^{λ_θ} the λ_θ-Reeb vector field, where $\lambda_\theta = \lambda_\phi|_{C_\theta}$, C_θ the fiber over θ of the projection $M \to S^1$. Analogously to Example 4.6 there is a (λ_ϕ, α)-admissible almost complex structure J^{λ_ϕ} on M with $J^{\lambda_\phi}(R^{\lambda_\theta}, 0) = (0, \partial_\theta)$. Indeed, let $\{J_\theta\}_{\theta\in\mathbb{R}}$ be a family of $d\lambda$-compatible almost complex structures on the contact distribution ξ of C, smooth in θ and constant near the integers, with $J_{\theta+1} = \phi_* J_\theta$; it exists as $\phi_* J_0$ is again $d\lambda$-compatible and the space of such structures is contractible. Then J_θ on $\xi \subset TC_\theta$, together with $J(R^{\lambda_\theta}, 0) = (0, \partial_\theta)$, descends to M, since ϕ preserves R^λ. It is ω-compatible, as $\omega = d\lambda$ on ξ and $\omega((R^{\lambda_\theta}, 0), (0, \partial_\theta)) = \lambda(R^{\lambda_\theta}) = 1$, and it preserves $\mathcal{D}$ and $\xi = \mathcal{D}^\perp$, hence it is admissible.

We now show that all Reeb 2-curves in M must be of a certain type. Let $o : S^1 \to C$ be a closed λ-Reeb orbit and suppose that $\mathrm{image}\, \phi^k \circ o = \mathrm{image}\, o$, for some $k > 0$, so that

$$\forall t \in S^1 : \phi^k(o(t)) = o(t + \theta_0)$$

for some $\theta_0 \in [0, 1)$, which is determined modulo $1/m(o)$, for $m(o)$ the multiplicity of o: there are exactly $m(o)$ choices of θ_0. Fix one. Let $\widetilde{o} : S^1 \times [0, k] \to C \times \mathbb{R}$ be the map

$$\widetilde{o}(t, \tau) = (o(t + \theta_0 \cdot \frac{\tau}{k}), \tau).$$

Then $\widetilde{o}$ is well defined on the quotient $T^2 \simeq S^1 \times ([0,k]/0 \sim k)$, and we denote the induced map by u^k_{o,θ_0}, called a *charge k generalized Reeb torus of o*. The $m(o)$ choices of θ_0 give $m(o)$ distinct maps with the same image, denoted $\mathcal{T}_o$: an embedded torus, the leaf through image $o \times \{0\}$ of the distribution $\mathcal{D}$, see the proof of Proposition 7.1. When $\phi^k \circ o = o$ we may take $\theta_0 = 0$, and then we write $u^k_o := u^k_{o,0}$; the other choices are the analogues of the sheared Reeb tori (6.3), to which they reduce for $\phi = \mathrm{id}$, $k = 1$. We denote by A^k_o the class of u^k_{o,θ_0} in $H_2(M,\mathbb{Z})$. It does not depend on θ_0: the $m(o)$ maps are coverings of the embedded torus $\mathcal{T}_o$ with the same ω-area, k times the period of o, hence of the same degree. For $\phi = \mathrm{id}$ and $\beta = [o]$ we have $A^k_o = k \cdot \beta \otimes [S^1] = A^k_\beta$, the class of Proposition 6.4, and as before we abbreviate A^1_β by A_β. In general A^k_o may depend on o and not only on k and $[o]$, which is why A is a free parameter in Proposition 7.1.

By construction, u^k_{o,θ_0} is a charge k Reeb 2-curve and its image is an embedded J^{λ_ϕ}-holomorphic torus $\mathcal{T}_o$. Moreover, u^k_{o,θ_0} is J^{λ_ϕ}-holomorphic with respect to a uniquely determined complex structure on T^2, similarly to the case of Reeb tori of Section 4.2. However, unlike the case of Reeb tori, this complex structure need not be "rectangular" when $\theta_0 \neq 0$: for J^{λ_ϕ} the lattice is $\mathbb{Z} + \mathbb{Z}(\theta_0 + \frac{ik}{c})$, for c the period of o.

Proposition 7.1. *Let $M = (M_{\phi,1}, \lambda_\phi, \alpha)$ be the mapping torus of a strict contactomorphism ϕ as above, and let J be a (λ_ϕ, α)-admissible almost complex structure on M. Then:*

(1) *Every Reeb 2-curve $u : \Sigma \to M$ maps each component of $\widetilde{\Sigma}$ on which it is not constant onto a torus $\mathcal{T}_o = \operatorname{image} u^k_{o,\theta_0}$, for an orbit string o with $\operatorname{image}\phi^k \circ o = \operatorname{image} o$, $k \geq 1$. If u has charge k, Σ being a nodal elliptic curve, its image is a single such torus, with $\operatorname{image}\phi^k \circ o = \operatorname{image} o$ for this k.*

(2) *For $k = 1$, $\beta \in \pi_1(C)$ and any $A \in H_2(M,\mathbb{Z})$, every element of $\overline{\mathcal{M}}^1_1(J, A, \iota_*\beta)$, for $\iota : C \to M$ the inclusion of a fiber, is represented by a J-holomorphic covering map $u : T^2 \to \mathcal{T}_o$ of charge 1, where o, with $\operatorname{image}\phi \circ o = \operatorname{image} o$, is the orbit string traced by u on any component of any regular fiber of $pr_{S^1} \circ u$, in the sense of Section 6.1, applied in the coordinates $C \times \{\theta\}$, $\theta \in [0,1)$, for $pr_{S^1} : M \to S^1$ the projection; all these components trace the same o, which is thus determined by u, and its class $\beta' \in \pi_1(C)$ satisfies $\iota_*\beta' = \iota_*\beta$. The curves tracing a given o are exactly the $m(o)$ generalized Reeb tori u^1_{o,θ_0}, one for each choice of θ_0; they all have class A^1_o, so they lie in $\overline{\mathcal{M}}^1_1(J, A, \iota_*\beta)$ iff $A = A^1_o$.*

(3) *The Fredholm index of the real linear CR operator (6.8) at u^k_{o,θ_0}, defined as there with $\mathcal{W}$ the pullback of ξ, u^k_{o,θ_0} being immersed, is 2, so that, subtracting the dimension of the translations, the expected dimension of $\overline{\mathcal{M}}^k_1(J, A^k_o, \beta)$ is 0, for β the π-class of u^k_{o,θ_0}.*

(4) *For $k = 1$, β and A as in part 2, the map*

$$\mathrm{Orb} : \overline{\mathcal{M}}^1_1(J, A, \iota_*\beta) \to \bigsqcup_{\beta'} \mathcal{O}(R^\lambda, \beta'), \quad u \mapsto o,$$

the union over the $\beta' \in \pi_1(C)$ with $\iota_*\beta' = \iota_*\beta$, with o as in part 2, is continuous and proper. Its fiber over o is empty unless $\operatorname{image}\phi \circ o = \operatorname{image} o$ and $A^1_o = A$, and then has exactly $m(o)$ elements; in particular Orb *is injective over simple orbit strings.*

Proof. The proof of part one is a variant of the proof of Proposition 6.4. Here $\mathcal{D} = \operatorname{span}(R^{\lambda_\theta}, \partial_\theta)$ is integrable, its leaves are the surfaces swept out by a Reeb trajectory under the flow of ∂_θ, and such a leaf is compact iff the trajectory is closed and ϕ^k preserves its image for some $k \geq 1$, that is iff the leaf is of the form $\mathcal{T}_o$: a leaf closed in M is embedded, pr_{S^1} restricts to it as a submersion, so its compact fibers are closed Reeb orbits in the fibers C_θ, and the ∂_θ-holonomy ϕ^{-1} permutes the finitely many components of its fiber over 0 cyclically. Let Σ_i be a component of $\widetilde{\Sigma}$ on which u is not constant. By the first condition of Definition 1.12 the rank of $d\widetilde{u}$ is 0 or 2 at each point of Σ_i, so on a component W

of $\{d\widetilde{u}\neq 0\}\cap\Sigma_i$ the map $\widetilde{u}$ is an immersion into a single leaf L, and $\widetilde{u}(W)$ is open in L. The frontier of $\widetilde{u}(W)$ in L lies in $\widetilde{u}(\overline{W}\setminus W)\subset\widetilde{u}(\{d\widetilde{u}=0\})$, which has Hausdorff dimension 0 by the Sard–Federer theorem [9, 3.4.3], as $\widetilde{u}$ is C^∞; a closed set of dimension 0 does not disconnect the surface L, so $\widetilde{u}(W)$ is open and closed in the complement of its frontier, that is $\widetilde{u}(W)=L\setminus F$ for F this frontier. The closure $\overline{L}$ of L in M lies in the compact set $\widetilde{u}(\overline{W})$, and $\overline{L}\setminus L\subset\widetilde{u}(\overline{W}\setminus W)$ has Hausdorff dimension 0; as the closure of a leaf is a union of leaves, each of dimension 2, $\overline{L}\setminus L=\emptyset$. So $L=\widetilde{u}(\overline{W})$ is compact, hence a torus $\mathcal{T}_{o_W}$.

There are finitely many such W. Fix Riemannian metrics on M and Σ_i. The area of $\widetilde{u}(W)$ is at most $\int_W|d\widetilde{u}|^2$, these integrals sum to at most $\int_{\Sigma_i}|d\widetilde{u}|^2<\infty$, and the area of a torus $\mathcal{T}_o$ is at least $c^{-1}\int_{\mathcal{T}_o}\omega$, which is at least c^{-1} times the minimal period of a closed R^λ-orbit, for c a bound on $|\omega|$. On the interior of $\{d\widetilde{u}=0\}$ the map $\widetilde{u}$ is locally constant, and each component of this interior has a frontier point in the closure of some W, so $\widetilde{u}(\Sigma_i)$ is contained in the finite union of the tori $\mathcal{T}_{o_W}$. Distinct leaves are disjoint, so the sets $\widetilde{u}^{-1}(\mathcal{T}_{o_W})\cap\Sigma_i$ are closed, pairwise disjoint or equal, and cover the connected Σ_i: they are all equal, and $\widetilde{u}(\Sigma_i)$ is a single torus $\mathcal{T}_o$.

The same argument on the connected Σ, whose constant components map into the tori of the adjacent ones, shows that a charge k curve u has image a single $\mathcal{T}_o$, and then image $\phi^k\circ o=$ image o: the charge of u is a multiple of the least $k'\geq 1$ with image $\phi^{k'}\circ o=$ image o, as the pullback of α to $\mathcal{T}_o$ integrates to 0 over o, and to k' over the loop obtained from the path $\tau\mapsto[(o(0),\tau)]$, $0\leq\tau\leq k'$, by closing up along o. For the second part, u is a Reeb 2-curve by Lemma 6.1, and its domain is a smooth torus, as in the proof of Proposition 6.4. By the first part u has image $\mathcal{T}_o$, which is an embedded J-holomorphic torus as it is tangent to $\mathcal{D}$, so u is a J-holomorphic map $T^2\to\mathcal{T}_o$, a covering map by the Riemann–Hurwitz formula.

As $k=1$, $\mathcal{T}_o\cap C_\theta=$ image o in the coordinates $C\times\{\theta\}$, so the components of a regular fiber of $pr_{S^1}\circ u$ over θ are the components of $u^{-1}($image $o\times\{\theta\})$, each covering image o with some degree. All have the same degree, the covering u being normal as $\pi_1(\mathcal{T}_o)$ is abelian, and they are positively oriented by the orientation argument in the proof of Proposition 6.4, which applies verbatim. So all components trace the same iterate of the simple orbit, and this is o. As in the proof of Proposition 6.4, the charge 1 coverings of $\mathcal{T}_o$ whose fibers trace o are classified by the index $m(o)$ subgroups of $\pi_1(\mathcal{T}_o)\simeq\mathbb{Z}^2$ surjecting onto π_1 of the base circle of $\mathcal{T}_o\to S^1$, of which there are $m(o)$, and the $m(o)$ maps u^1_{o,θ_0} realize them, their subgroups being pairwise distinct. All of these have class A^1_o, by the definition of A^1_o.

We prove Part 3. First $c_1(A^k_o)=0$. As u^k_{o,θ_0} covers $\mathcal{T}_o$, it suffices that $\langle c_1(TM,J),[\mathcal{T}_o]\rangle=0$. Now $TM|_{\mathcal{T}_o}=T\mathcal{T}_o\oplus\xi|_{\mathcal{T}_o}$ as complex vector bundles, J preserving $\mathcal{D}=T\mathcal{T}_o$ and $\mathcal{D}^\perp=\xi$, and $c_1(T\mathcal{T}_o)=\chi(T^2)=0$. The symplectic bundle $(\xi|_{\mathcal{T}_o},d\lambda)$ has a flat symplectic connection: parallel transport along ∂_θ is the identity in the coordinates $C\times\mathbb{R}$, and along R^λ it is the linearized Reeb flow, cf. the proof of Proposition 6.7; these commute as $[\partial_\theta,R^\lambda]=0$ on $C\times\mathbb{R}$, and the connection descends to M as $\phi_*R^\lambda=R^\lambda$, so it is flat. A flat symplectic bundle over T^2 with commuting holonomies $A,B\in Sp(2n,\mathbb{R})$ has $c_1=0$: for b a path from id to B, trivializing over the cylinder by b exhibits the bundle as obtained by clutching with the loop $\gamma(y)=b(y)^{-1}A\,b(y)$, so c_1 is the Maslov index of γ, which vanishes since the map $\rho:Sp(2n,\mathbb{R})\to S^1$ of [27, Theorem 2.29], see also Gutt [14, Proposition 6, (2)], is conjugation invariant, hence constant on γ. By the Riemann–Roch theorem the index of (6.8) is then $(n+1)\chi(T^2)+2c_1(A^k_o)+2=2$, where $\dim C=2n+1$ and the summand 2 is $\dim T_j\mathcal{T}$.

The last part follows from the first two: the fiber of Orb over o consists of the charge 1 curves of class A tracing o, which by part 2 are the $m(o)$ generalized Reeb tori u^1_{o,θ_0} if image $\phi\circ o=$ image o and $A=A^1_o$, and there are none otherwise; Orb is continuous, by the argument of Lemma 6.5, and it is proper since the curves over a compact set of orbit strings have bounded energy, so that the argument given for $\widetilde{\mathrm{Orb}}$ in the proof of Theorem 1.8 applies. □

Proof of Theorem 1.14. Let $u : \Sigma \to (M = M_{\widetilde{\phi},1})$ be a Reeb 2-curve in the mapping torus as in the statement. By part one of Proposition 7.1, the image of u contains a torus $\mathcal{T}_o$ with image $\widetilde{\phi}^k \circ o = \text{image}\, o$ for some $k \geq 1$, u being non-constant by the second condition of Definition 1.12. That is, $\widetilde{\phi}^k$ fixes the image of some closed Reeb orbit, so that ϕ^k fixes the image of some closed g_{st}-geodesic, which is impossible by assumption. □

Proposition 7.2. *Let (C, λ) be a closed contact manifold with λ satisfying one of the following conditions:*

(1) *There is a non-degenerate λ-Reeb orbit.*

(2) *$i(N, R^\lambda, \beta) \neq 0$ for some open compact $N \subset \mathcal{O}(R^\lambda, \beta)$, and some β.*

Then:

(1) *Let (λ, α) be the lcs-fication of (C, λ). There exists an $\epsilon > 0$ s.t. for any tamed lcs pair (λ', α', J) on $M = C \times S^1$, with $(d_{\alpha'}\lambda', J)$ ϵ-close to $(d_\alpha \lambda, J^\lambda)$ (as in Definition 5.9), there exists an elliptic, J-holomorphic α-charge 1 curve u in M.*

(2) *In addition, if (λ', α') is first kind and M has dimension 4 then u may be assumed to be normal and embedded.*

Proof. If we have a closed non-degenerate λ-Reeb orbit o then we also have an open compact subset $N = \{o\} \subset \mathcal{O}(R^\lambda, \beta)$, with $i(N, R^\lambda, \beta) = i(o)/m(o) = \pm 1/m(o) \neq 0$ by (A.1). Thus suppose that the condition 2 holds.

Set

$$\widetilde{N} := \mathcal{P}(N) \subset \overline{\mathcal{M}}_1^1(J^\lambda, A_\beta, \beta),$$

which is an open compact set of Reeb tori, by Proposition 6.4, Lemma 6.5 and the properness of Orb, (6.6). By Theorem 6.14, and by the assumption that $i(N, R^\lambda, \beta) \neq 0$,

$$GW_1^1(\widetilde{N}, J^\lambda, A_\beta, \beta) \neq 0.$$

The first part of the conclusion then follows by Proposition 5.14.

We now verify the second part. Suppose that M has dimension 4. Let (λ', α', J) be as in the first part, with (λ', α') first kind, and let $u \in \overline{\mathcal{M}}_1^1(J, A_\beta, \beta)$, which exists by the proof of the first part.

For convenience, we now recall the adjunction inequality.

Theorem (McDuff–Micallef–White [29], [24])**.** *Let (M, J) be an almost complex 4-manifold and let $A \in H_2(M)$ be a homology class that is represented by a simple J-holomorphic curve $u : \Sigma \to M$. Then $c_1(A) \leq A \cdot A + \chi(\Sigma)$, with equality if and only if u is an embedding,* [28, Theorem 2.6.4].

First, u is normal: otherwise $\widetilde{\Sigma}$ has a spherical component, which is impossible as in the proof of Proposition 6.4. Let $\underline{u}$ be a simple J-holomorphic curve covered by u, see for instance [28, Proposition 2.5.1]. Its domain Σ' is a closed Riemann surface, and $u = \underline{u} \circ \pi$ for a holomorphic branched covering $\pi : T^2 \to \Sigma'$ of its domain; Σ' is not a sphere, as in the previous sentence, so by the Riemann–Hurwitz formula $\Sigma' \simeq T^2$ and π is a covering. The charge of u is a positive multiple of the charge of $\underline{u}$, so $\underline{u}$ has charge 1. Now apply the adjunction inequality to $\underline{u}$, of class A' with $A_\beta = (\deg \pi) A'$: $c_1(A') = c_1(A_\beta)/\deg \pi = 0$, as J is C^∞-close, hence homotopic, to J^λ, so that $c_1(A_\beta) = 0$ by Section 4.1, $A' \cdot A' = 0$, as A_β is represented by $\gamma \times S^1$ for a loop γ in the 3-manifold C, which is disjoint from a push-off of itself, and $\chi(\Sigma') = 0$, so equality holds and $\underline{u}$ is an embedding. So $\underline{u}$ is an embedded, normal, charge 1 elliptic J-curve, which is our claim.

□

Proof of Theorem 3.3. Let $\mathcal{X} \ni (\lambda, \alpha, J_0 := J^\lambda)$ be the set of triples (λ', α', J) with $(\lambda', \alpha') \in \mathcal{F}(M)$, J (λ', α')-admissible, and $(d_{\alpha'}\lambda', J)$ ϵ-close to $(\omega_0 := d_\alpha \lambda, J_0)$ in the sense of Definition 5.9, for ϵ as in the first part of Proposition 7.2. To prove the theorem we need to construct a map $E : V \to \mathcal{J}(M)$, where V is some neighborhood of (λ, α), the U of the theorem, in the space $(\mathcal{F}(M), d_\infty)$ (see Definition 3.1) and where

$$\forall (\lambda', \alpha') \in V : (\lambda', \alpha', E(\lambda', \alpha')) \in \mathcal{X}.$$

Proposition 7.2 then gives, for each $(\lambda', \alpha') \in V$, an $E(\lambda', \alpha')$-holomorphic elliptic curve u of α-charge 1 in M. Using Lemma 6.1 we would then conclude that there is an elliptic Reeb 2-curve u in (M, λ', α'). If M has dimension 4 then in addition u may be assumed to be normal and embedded, and symplectic, as $E(\lambda', \alpha')$ is ω-compatible. If α' is integral then $[\alpha'] = [\alpha]$ for V small, as the periods of α' over fixed loops generating $H_1(M, \mathbb{Z})$ are integers close to those of α. So the α'-charge of u is its α-charge, 1. And so we will be done.

Let $(\lambda', \alpha') \in \mathcal{F}(M)$ and write $\omega = d_{\alpha'}\lambda'$. Then $\mathcal{V}_{\lambda'}$ and $\xi_{\lambda'}$ are smooth distributions, the pair being first kind, and $TM = \mathcal{V}_{\lambda'} \oplus \xi_{\lambda'}$. Define $E(\lambda', \alpha')$ to be $\mathrm{ret}(g|_{\mathcal{V}_{\lambda'}}, \omega|_{\mathcal{V}_{\lambda'}}) \oplus \mathrm{ret}(g|_{\xi_{\lambda'}}, \omega|_{\xi_{\lambda'}})$ on $TM = \mathcal{V}_{\lambda'} \oplus \xi_{\lambda'}$, for ret the retraction of the proof of Lemma 5.13 applied fiberwise, and $g := \omega_0(\cdot, J_0 \cdot)$. It is (λ', α')-admissible, as the summands are ω-orthogonal; $E(\lambda, \alpha) = J_0$, as $g = g_{J_0}$ and J_0 preserves the splitting; and it depends continuously on (λ', α') in the C^∞ topology, $\mathcal{V}_{\lambda'} = \ker d\lambda'$ having constant rank. So if we take V to be the δ-ball in $(\mathcal{F}(M), d_\infty)$ centered at (λ, α), for δ sufficiently small, then it has the needed property.

□

Proof of Theorem 1.8. Let (C, λ) and ϕ be as in the hypothesis. Let $\{\phi_t\}$, $t \in [0, 1]$ be a smooth family of strict contactomorphisms with $\phi_0 = \mathrm{id}$, $\phi_1 = \phi$. This gives a smooth fibration $\widehat{M} := C \times \mathbb{R} \times [0, 1]/((x, \theta, t) \sim (\phi_t(x), \theta + 1, t)) \to [0, 1]$, with fiber over $t \in [0, 1]$: $M_{\phi_t, 1}$, which is moreover endowed with the first kind lcs pair $(\lambda_{\phi_t}, \alpha)$, where this is the "mapping torus structure" as in Section 7.1. Let $tr : \widehat{M} \to (C \times S^1) \times [0, 1]$ be a smooth trivialization covering the identity of $S^1 \times [0, 1]$, for instance $tr_t[(x, \theta)] := (\phi^{-1}_{t\mu(\theta)}(x), \theta)$, $\theta \in [0, 1]$, with $\mu : [0, 1] \to [0, 1]$ smooth, 0 near 0 and 1 near 1, which is well defined as $(x, 0) \sim (\phi_t(x), 1)$ and both map to $(x, 0)$, and restricting to the identity $C \times S^1 \to C \times S^1$ over 0. Pushing forward by the bundle map tr, the above mentioned family of lcs pairs, we get a smooth family $\{(\lambda_t, \alpha)\}$, $t \in [0, 1]$, of first kind integral lcs pairs on $C \times S^1$, with $(\lambda_0, \alpha) = (\lambda, \alpha)$ the standard lcs-fication of λ.

Fix a smooth family $\{J^{\lambda_t}\}$ of almost complex structures on $C \times S^1$ with each J^{λ_t} admissible with respect to (λ_t, α), and with $J^{\lambda_0} = J^\lambda$ the almost complex structure of Example 4.6; it exists as the admissible structures for each pair form a contractible space, see the remark following Definition 4.3. Let $N \subset \mathcal{O}(R^\lambda, \beta)$ be an open compact set satisfying $i(N, R^\lambda, \beta) \neq 0$.

We first reduce to the case $\hat{i}(N, R^\lambda, \beta) \neq 0$, for $\hat{i}$ as in (6.12). As noted there, $N^{(e)} = \emptyset$ for e large, so there is a largest e_0 with $i(N^{(e_0)}, R^\lambda) \neq 0$, and $e_0 \geq 1$ as $N^{(1)} = N$. Set $N' := N^{(e_0)}$, an open compact set of orbit strings meeting finitely many classes γ, each with $e_0\gamma = \beta$. Since $(N')^{(e)} = N^{(e_0 e)}$, for $(N')^{(e)} := \{o'' \,|\, e\, o'' \in N'\} = \bigsqcup_\gamma (N' \cap \mathcal{O}(R^\lambda, \gamma))^{(e)}$, has vanishing Fuller index for $e \geq 2$, (6.12) gives $\sum_\gamma \hat{i}(N' \cap \mathcal{O}(R^\lambda, \gamma), R^\lambda, \gamma) = i(N', R^\lambda) \neq 0$, so that $\hat{i}(N' \cap \mathcal{O}(R^\lambda, \gamma), R^\lambda, \gamma) \neq 0$ for some γ. If $o' \in N'$ is a fixed Reeb string of ϕ then so is its e_0-fold iterate $e_0\, o' \in N$, which is in class β. Replacing (N, β) by $(N' \cap \mathcal{O}(R^\lambda, \gamma), \gamma)$ we may therefore assume that $\hat{i}(N, R^\lambda, \beta) \neq 0$.

For each t, part 4 of Proposition 7.1, applied to $M_{\phi_t, 1}$ with the class $tr_{t*}^{-1} A_\beta$ and transported by tr_t, gives a continuous proper map $\mathrm{Orb} : \overline{\mathcal{M}}^1_1(J^{\lambda_t}, A_\beta, \beta) \to \mathcal{O}(R^\lambda, \beta)$, the traced string having class β in C since the fiber inclusion is injective on free homotopy classes for $C \times S^1$ and tr_t restricted to each fiber is isotopic to the identity via $s \mapsto tr_s$; these induce a continuous proper map

$$\widetilde{\mathrm{Orb}} : \widetilde{\mathcal{M}} = \overline{\mathcal{M}}^1_1(\{J^{\lambda_t}\}, A_\beta, \beta) \to \mathcal{O}(R^\lambda, \beta) \times [0, 1],$$

defined by $\widetilde{\mathrm{Orb}}(u,t) = (\mathrm{Orb}(u),t)$. It is continuous by the argument of Lemma 6.5, the fibers of $pr_{S^1}\circ u_i$ converging to those of $pr_{S^1}\circ u$ when $(u_i,t_i)\to(u,t)$, and proper: the curves over a compact set of pairs (o,t) have energy bounded by the periods of the o, so a sequence of them has a Gromov convergent subsequence by Theorem 5.2, whose limit lies in $\widetilde{\mathcal{M}}$ by Lemma 4.9. Set

$$\widetilde{N} = \widetilde{\mathrm{Orb}}^{-1}(N\times[0,1]).$$

As N is open and compact in $\mathcal{O}(R^\lambda,\beta)$ and $\widetilde{\mathrm{Orb}}$ is continuous and proper, $\widetilde{N}$ is an open and compact subset of $\widetilde{\mathcal{M}}$. Set $N_i := \{u\,|\,(u,i)\in\widetilde{N}\}$ for $i=0,1$, so that by construction $N_0 = \mathrm{Orb}^{-1}(N)$, for Orb as in (6.6) for the lcs-fication (λ_0,α). We use $\mathrm{Orb}^{-1}(N)$ rather than the set of Reeb tori $\mathcal{P}(N)$: for $t>0$ the pair (λ_t,α) is not an lcs-fication, so Lemma 6.5 does not apply to the family $\{J^{\lambda_t}\}$, and $\mathcal{P}(N)$ need not be the slice at $t=0$ of an open and closed subset of $\widetilde{\mathcal{M}}$.

It follows by Theorem 6.14 that

$$GW_1^1(N_0,J^{\lambda_0},A_\beta,\beta) = \hat{i}(N,R^\lambda,\beta)\neq 0.$$

Then applying Lemma 5.6, as $c_1(A_\beta)=0$ by Section 4.1, we get that

$$GW_1^1(N_1,J^{\lambda_1},A_\beta,\beta)\neq 0.$$

So $N_1\neq\emptyset$, and an element of N_1 is the transport by tr_1 of a generalized Reeb torus u^1_{o,θ_0} of $M_{\phi,1}$ with $o=\mathrm{Orb}(u^1_{o,\theta_0})\in N$, by construction of $\widetilde{N}$ and Proposition 7.1. In particular image $\phi\circ o=$ image o, so $\phi\circ o=\theta_0\cdot o$ for some θ_0, as in Section 7.1: $o\in N$ is a class β fixed Reeb string of ϕ, Definition 1.1; for the original (N,β) take the e_0-fold iterate, as noted above. □

Proof of Theorem 1.7. Suppose that λ is Morse–Bott and we have a connected component $N\subset\mathcal{O}(R^\lambda,\beta)$, which is open and compact by Appendix A, with O_N orientable and with $\chi^{orb}(N)\neq 0$, so that $i(N,R^\lambda,\beta)\neq 0$ by (A.3). If λ' is sufficiently C^1 nearby to λ then we may find an open compact $N'\subset\mathcal{O}(R^{\lambda'},\beta)$ with

$$i(N',R^{\lambda'},\beta) = i(N,R^\lambda,\beta)\neq 0.$$

Indeed, let $\lambda_t := (1-t)\lambda + t\lambda'$; for λ' C^1-close each λ_t is contact and R^{λ_t} is C^0-close to R^λ, so by [38, Lemma 4.1] and the Arzelà–Ascoli argument of Appendix A there is an open compact $\widetilde{N}\subset S(\{R^{\lambda_t}\},\beta)/S^1$ with slice N at $t=0$; let N' be its slice at $t=1$ and use the invariance of the Fuller index, Appendix A. Then the result follows by Theorem 1.8. □

Proof of Theorem 1.2. The form λ_{st} is Morse–Bott, see [4], and the simple Hopf fibers form a connected component $N\simeq\mathbb{CP}^n$ of $\mathcal{O}(R^{\lambda_{st}},0)$, with $O_N = S^{2n+1}$ in the notation of Appendix A, orientable, all of whose orbits are simple, so $m\equiv 1$ on N, so that $\chi^{orb}(N)=\chi(\mathbb{CP}^n)=n+1\neq 0$. Apply Theorem 1.7. □

Proof of Theorem 3.4. Let $u:\Sigma\to M$ be an immersed Reeb 2-curve and $\widetilde{u}:\widetilde{\Sigma}\to M$ its normalization, an immersion with $d\widetilde{u}(T\widetilde{\Sigma})=\mathcal{D}$. Set $\eta := \widetilde{u}^*\lambda$. By the Reeb condition $\lambda(X_\alpha)>0$, while $\lambda(X_\lambda) = \omega(X_\lambda,X_\lambda)=0$, so λ restricts to a non-zero linear form on each $\mathcal{D}(p)$, and η is a nowhere vanishing 1-form on the closed surface $\widetilde{\Sigma}$. In particular every component of $\widetilde{\Sigma}$ has Euler characteristic 0. The foliation $\ker\eta$ of $\widetilde{\Sigma}$ admits a closed transversal $\sigma:S^1\to\widetilde{\Sigma}$ with $\eta(\sigma')>0$. Indeed, $\ker\eta$ is transversely oriented by η, so it is tangent to a nowhere vanishing vector field V on the oriented surface $\widetilde{\Sigma}$. If some orbit of V is not closed, take a point q in its limit set, in the dynamical sense, and a short arc τ through q transverse to V; the orbit crosses τ at infinitely many distinct points, and an orbit segment between two consecutive crossings, closed up by the sub-arc of τ between them, can be perturbed to a closed transversal, see for instance [5, Proposition 1.17]. If every orbit of V is closed, the first return map to a transverse arc at any orbit is an orientation preserving homeomorphism of an interval all of whose points are periodic, hence the identity; so the orbits are the fibers of a circle bundle over a circle, and

a section of this bundle is a closed transversal. Along it η does not vanish, so $\eta(\sigma') > 0$ after possibly reversing the orientation of σ. Then $o = \widetilde{u} \circ \sigma$ satisfies $\lambda(o') > 0$ and $o' \in \mathcal{D}$, so it is a Reeb 1-curve.

For the second part, if an immersed elliptic Reeb 2-curve u is not normal, then $\widetilde{\Sigma}$ has a genus 0 component, contradicting the above. □

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A. Fuller index

Let X be a complete vector field without zeros on a smooth manifold M. Set

$$S(X,\beta) = \{o \in L_\beta M \mid \exists p \in (0,\infty),\ \ o : \mathbb{R}/\mathbb{Z} \to M \text{ is a periodic orbit of } pX\},$$

where $L_\beta M$ denotes the free homotopy class β component of the free loop space

$$LM = \{o : S^1 \to M \,|\, o \text{ is smooth}\},$$

with the compact open topology, where $S^1 = \mathbb{R}/\mathbb{Z}$. The above p is uniquely determined and we denote it by $p(o)$ called the period of o.

There is a natural S^1 reparametrization action on $S(X,\beta)$: $t \cdot o$ is the loop $t \cdot o(\tau) = o(t+\tau)$. The elements of $\mathcal{O}(X,\beta) := S(X,\beta)/S^1$ will be called *orbit strings*. Slightly abusing notation we just write o for the equivalence class of o.

An orbit string is ***simple*** if its parametrization by $\mathbb{R}/\mathbb{Z}$ is injective; every orbit string o covers a unique simple one, o_s. The multiplicity $m(o)$ of an orbit string is the ratio $m(o) := p(o)/p(o_s) \in \mathbb{Z}_{\geq 1}$. We write $d\,o$ for the d-fold iterate of o, $d \geq 1$, $(d\,o)(\tau) := o(d\tau)$, of period $d\,p(o)$, so that $o = m(o)\,o_s$ for the simple orbit o_s covered by o.

We want a kind of fixed point index of an open compact subset $N \subset \mathcal{O}(X,\beta)$, which counts orbit strings o with certain weights. Suppose first that $N \subset \mathcal{O}(X,\beta)$ is finite; each $o \in N$ is then isolated in $\mathcal{O}(X,\beta)$, as N is open and $\mathcal{O}(X,\beta)$ is Hausdorff. Then to such an (N,X,β) Fuller [13] associates an index:

$$i(N,X,\beta) = \sum_{o \in N} \frac{1}{m(o)} i(o),$$

where $i(o)$ is the fixed point index at $o(0)$ of the Poincaré return map, with return time near $p(o)$, of a local hypersurface of section in M through $o(0)$ transverse to the image of o.

Fuller then shows that $i(N,X,\beta)$ has the following invariance property. Here a homotopy $\{X_t\}$ of vector fields is continuous in t for the C^1 topology, and each X_t is complete without zeros. For a continuous homotopy $\{X_t\}$, $t \in [0,1]$ set

$$S(\{X_t\},\beta) = \{(o,t) \in L_\beta M \times [0,1] \,|\, o \in S(X_t,\beta)\}.$$

Fuller does not fix the class β, but $S(\{X_t\},\beta)/S^1$ is open and closed in the space of all orbit strings of the homotopy, the class being locally constant. Given a continuous homotopy $\{X_t\}$, $X_0 = X$, $t \in [0,1]$, suppose that $\widetilde{N}$ is an open compact subset of $S(\{X_t\},\beta)/S^1$, such that

$$\widetilde{N} \cap (L_\beta M \times \{0\})\,/S^1 = N.$$

Then if

$$N_1 = \widetilde{N} \cap (L_\beta M \times \{1\})\,/S^1$$

is also finite, we have

$$i(N,X,\beta) = i(N_1,X_1,\beta).$$

For a general open compact $N \subset \mathcal{O}(X,\beta)$ the index is defined by perturbation. There is a C^1-small homotopy $\{X_t\}$, $X_0 = X$, and an open compact $\widetilde{N}$ as above with N_1 finite: the class β orbits of a generic X_1 with period in a bounded range are non-degenerate, that is their linearized return maps have no eigenvalue 1 (Kupka–Smale), hence isolated, so that the compact N_1 is finite; the perturbation is supported near the compact union of the images of the orbits in N, and $\widetilde{N}$ exists by [38, Lemma 4.1] and the Arzelà–Ascoli theorem. We set $i(N,X,\beta) := i(N_1,X_1,\beta)$. By the invariance property for finite N, N_1, applied to the concatenation of the reverse of one homotopy with the other, with the union of the two sets $\widetilde{N}$, glued along N, as the open compact set, this is independent of the choices; and the invariance property then holds for general N, N_1, see Fuller [13].

In the case where X is the R^λ-Reeb vector field on a contact manifold (C^{2n+1},λ), and if o is non-degenerate, that is 1 is not an eigenvalue of $F^\lambda_{p(o),*}$ on $\xi = \ker\lambda$, we have:

$$i(o) = \operatorname{sign}\operatorname{Det}(\operatorname{id}|_{\xi(x)} - F^\lambda_{p(o),*}|_{\xi(x)}) = (-1)^{CZ(o)-n}, \tag{A.1}$$

where $x = o(0)$, $F^\lambda_{p(o),*}$ is the differential at x of the time $p(o)$ flow map of R^λ, and where $CZ(o)$ is the Conley–Zehnder index of the path $t \mapsto F^\lambda_{t,*}|_{\xi(x)}$, $0 \le t \le p(o)$, in a trivialization of ξ along o; the parity of $CZ(o)$ does not depend on the trivialization, see [33]; the second equality is the determinant property of the Conley–Zehnder index, Salamon [35], see also Gutt [14, Proposition 8, (7)].

The Morse–Bott case. The ***action spectrum*** of λ is the image of the period map $\operatorname{period} : S(R^\lambda) \to (0,\infty)$, $S(R^\lambda) := \bigcup_\beta S(R^\lambda,\beta)$, $o \mapsto \int_{S^1} o^*\lambda = p(o)$; we say that it is ***discrete*** if it is a discrete subset of $\mathbb{R}$. Recall [4] that a contact form λ on a closed manifold C^{2n+1} is *Morse–Bott* if its action spectrum is discrete, and if for every a in the action spectrum

$$P_a := \{x \in C \mid F^\lambda_a(x) = x\},$$

for F^λ_a the time a flow map of R^λ, is a closed submanifold of C with $T_xP_a = \ker(F^\lambda_{a,*} - \operatorname{id})_x$ and with $\operatorname{rank} d\lambda|_{P_a}$ locally constant. Suppose that λ is Morse–Bott and let $N \subset \mathcal{O}(R^\lambda,\beta)$ be a connected component. As the period map is continuous with discrete image, all $o \in N$ have the same period a, and $N = O_N/S^1$, where $O_N \subset P_a$ is the union of the images of the orbits in N, which is a connected component of P_a (the free homotopy class of the period a orbit through $x \in P_a$ is locally constant in x, and O_N is connected, being the preimage of N under the quotient by a connected group), and where the circle $\mathbb{R}/a\mathbb{Z}$, identified with S^1 by $t \mapsto t/a$, acts on O_N by the flow of R^λ. We assume that O_N is orientable; this holds for instance when $O_N = C$. This action is locally free, the isotropy group of $x \in O_N$ being cyclic of order the multiplicity of the element of N through x. Thus N is a closed orbifold, in general non-effective, of dimension $\dim O_N - 1$, and we write $m(q)$ for the order of the isotropy group of $q \in N$, which is the multiplicity of the corresponding orbit string. The *orbifold Euler characteristic* of N is

$$\chi^{orb}(N) := \sum_\sigma \frac{(-1)^{\dim\sigma}}{m(\sigma)},$$

the sum over the open cells σ of a cell decomposition of N along each of whose cells m is constant, equal to $m(\sigma)$, which exists as N is the orbit space of a compact group action, see Satake [37]; the sum is $\sum_S \chi_c(S)/m_S$ over the level sets $S = m^{-1}(m_S)$, for χ_c the Euler characteristic with compact supports, so it does not depend on the decomposition. When m is constant on N, N is a closed manifold and $\chi^{orb}(N) = \chi(N)/m$. In general $\chi^{orb}(N)$ and $\chi(N)$ need not vanish simultaneously: for a 2-sphere with four cone points of order 2, $\chi^{orb} = 0$ while $\chi = 2$.

We now compute $i(N,R^\lambda,\beta)$, following Bourgeois [4, Section 2.2], cf. [38, Section 2.1.1]. Choose $\delta > 0$ so that a is the only point of the action spectrum in $[a-\delta, a+\delta]$, and let U be a small tubular neighborhood of O_N with $\overline{U} \cap P_a = O_N$, so that the closed R^λ-orbits with period in $[a-\delta,a+\delta]$ meeting $\overline{U}$ are exactly the elements of N. Let $\bar f : N \to \mathbb{R}$ be a Morse function on the orbifold N, meaning that its pullback F to O_N is Morse–Bott with critical set a finite union of S^1-orbits; the critical points of $\bar f$ are the images of these orbits, and $\operatorname{ind}_{\bar f}(q)$ is the Morse–Bott index of F along the

orbit over q (such $\bar f$ exist, see [4, Section 2.2]). Extend the pullback of $\bar f$ to a smooth function f on C, supported in U and constant near O_N along the fibers of a tubular projection $U \to O_N$, so that in particular $df(R^\lambda) = 0$ on O_N, and set

$$\lambda_\mu := (1 + \mu f)\lambda, \quad \mu \geq 0 \text{ small.}$$

For a critical point q of $\bar f$ let o_q be the R^λ-orbit string over q. Along the image of o_q we have $df = 0$, as f is constant along the fibers of the tubular projection and its restriction to O_N is critical there, so that $R^{\lambda_\mu} = (1 + \mu f)^{-1} R^\lambda$ along the image of o_q. Consequently o_q is a closed R^{λ_μ}-orbit of period $(1 + \mu \bar f(q))a$ with the same image, hence with the same multiplicity $m(o_q) = m(q)$ and in the same free homotopy class β. By the proofs of [4, Lemmas 2.3, 2.4], which are local near O_N and use only that f is constant along the fibers near O_N and that $\bar f$ is Morse, for $\mu > 0$ sufficiently small these are all of the closed R^{λ_μ}-orbits contained in U with period in $(a - \delta, a + \delta)$, such orbits lying in any given neighborhood of O_N for μ small by the Arzelà–Ascoli argument below, a limit orbit having period a and lying in $\overline{U} \cap P_a = O_N$, they are non-degenerate, and

$$CZ(o_q) = CZ(N) - \tfrac{1}{2} \dim N + \operatorname{ind}_{\bar f}(q), \tag{A.2}$$

where $\operatorname{ind}_{\bar f}(q)$ is the Morse index of $\bar f$ at q, and $CZ(N)$ is the generalized Conley–Zehnder index [33] of the degenerate orbits in N, which is constant on N. (As usual $CZ(N)$ depends on the choice of a trivialization of the contact distribution along the orbits, but $CZ(N) - \frac{1}{2} \dim N$ is an integer whose parity does not.)

By an Arzelà–Ascoli argument, after shrinking μ if necessary, no closed $R^{\lambda_{t\mu}}$-orbit, $t \in [0, 1]$, with period in $[a - \delta, a + \delta]$ meets ∂U or has period $a \pm \delta$. Consequently the pairs (o, t), $t \in [0, 1]$, with o a class β orbit string of $R^{\lambda_{t\mu}}$ with image in U and period in $(a - \delta, a + \delta)$, form an open compact subset $\widetilde N$ of $S(\{R^{\lambda_{t\mu}}\}, \beta)/S^1$ with $\widetilde N \cap (L_\beta C \times \{0\})/S^1 = N$ and, by the above,

$$\widetilde N \cap (L_\beta C \times \{1\})/S^1 = \{o_q\}_{q \in \operatorname{Crit} \bar f}.$$

In particular N, the slice at $t = 0$, is open and compact in $\mathcal{O}(R^\lambda, \beta)$. So by the invariance of the Fuller index, (A.1) and (A.2):

$$i(N, R^\lambda, \beta) = \sum_{q \in \operatorname{Crit} \bar f} \frac{(-1)^{CZ(o_q) - n}}{m(o_q)} = (-1)^{CZ(N) - \frac{1}{2} \dim N - n} \sum_{q \in \operatorname{Crit} \bar f} \frac{(-1)^{\operatorname{ind}_{\bar f}(q)}}{m(q)} = \pm \chi^{orb}(N). \tag{A.3}$$

The last equality is the Morse theoretic computation of the orbifold Euler characteristic, that is the Poincaré–Hopf theorem for orbifolds [37] applied to the gradient of $\bar f$; when m is constant on N it is just the classical Morse count of $\chi(N)$, divided by m. Satake's theorem is for effective orbifolds whose singular set has codimension at least two. As S^1 is connected it preserves the orientation of O_N, so the isotropy group of $x \in O_N$ acts on the slice $T_x O_N / \mathbb{R} R^\lambda$ by orientation preserving linear maps of finite order, whose fixed subspaces have even codimension; so the second condition holds once the first does. Also N is orientable, the slices being oriented by the orientation of O_N and R^λ, compatibly with the local groups. We reduce to the effective case.

A finite subgroup of S^1 fixing an open subset of O_N pointwise fixes all of O_N: the fixed point set of a diffeomorphism of finite order is a closed submanifold, whose full dimensional part is then open and closed, and O_N is connected. Let $m_0 := \min m$. The set $\{m = m_0\}$ is open, as by the slice theorem the isotropy group of a point near $x \in O_N$ is a subgroup of that of x. So $\mathbb{Z}_{m_0} \subset S^1$ acts trivially on O_N. Hence m_0 divides every $m(q)$, and $S^1/\mathbb{Z}_{m_0}$ acts on O_N with isotropy groups of orders m/m_0, which act effectively on the slices, since an element acting trivially on a slice fixes a neighborhood of the point, its action on the slice being linearizable, hence all of O_N, so it lies in $\mathbb{Z}_{m_0}$. Thus N, with the orbifold structure given by this action, is effective, its orbifold Euler characteristic is $m_0 \chi^{orb}(N)$, and its Morse count for $\bar f$ is m_0 times $\sum_q (-1)^{\operatorname{ind}_{\bar f}(q)}/m(q)$. So the last equality for this structure implies it for N. In particular $i(N, R^\lambda, \beta) \neq 0$ if and only if $\chi^{orb}(N) \neq 0$.

B. Remark on multiplicity

This is a small note on how one deals with curves having non-trivial isotropy groups, in the virtual fundamental class technology. We primarily need this for the proof of Theorem 6.14.

Given a closed oriented orbifold $\mathfrak{X}$, with an orbibundle E over $\mathfrak{X}$, Fukaya–Ono [12] show how to construct using multi-sections its rational homology Euler class, which when $\mathfrak{X}$ represents the moduli space of some stable curves, is the virtual moduli cycle $[\mathfrak{X}]^{vir}$. When this is in degree 0, the corresponding Gromov–Witten invariant is $\int_{[\mathfrak{X}]^{vir}} 1$. However, they assume that their orbifolds are effective. This assumption is not really necessary for the purpose of construction of the Euler class but is convenient for other technical reasons. A different approach to the virtual fundamental class which emphasizes branched manifolds is used by McDuff–Wehrheim, see for example McDuff [23], [26] which does not have the effectivity assumption; a similar use of branched manifolds appears in [7]. In the case of a non-effective orbibundle $E \to \mathfrak{X}$ McDuff [25], constructs a homological Euler class $e(E)$ using multi-sections, which extends the construction [12]. McDuff shows that this class $e(E)$ is Poincaré dual to the completely formally natural cohomological Euler class of E, constructed by other authors. In other words there is a natural notion of a homological Euler class of a possibly non-effective orbibundle. We shall assume the following black box property of the virtual fundamental class technology.

Axiom B.1. *Suppose that the moduli space of stable maps is cleanly cut out, which means that it is represented by a (possibly non-effective) orbifold $\mathfrak{X}$ with an orbifold obstruction bundle E, that is the bundle over $\mathfrak{X}$ of cokernel spaces of the linearized CR operators. Then the virtual fundamental class $[\mathfrak{X}]^{vir}$ coincides with $e(E)$.*

Given this axiom it does not matter to us which virtual moduli cycle technique we use. It is satisfied automatically by the construction of McDuff–Wehrheim (at the moment in genus 0). It can be shown to be satisfied in the approach of John Pardon [32], cf. [32, Section 5.3] for the case of a manifold with obstruction bundle. And it is satisfied by the construction of Fukaya–Oh–Ohta–Ono [10], the latter was communicated to me by Kaoru Ono. When $\mathfrak{X}$ is 0-dimensional this does follow immediately by the construction in [12], taking at each isolated point the Kuranishi neighborhood $(V \times W, E \oplus W, \Gamma)$, with Γ acting diagonally and section $(v, w) \mapsto (s(v), w)$, for W a faithful Γ-representation; it is effective, with zero set $s^{-1}(0) \times \{0\}$ (this suffices for our paper).

As a special case most relevant to us here, suppose we have a moduli space of elliptic curves in M, which is regular with expected dimension 0, and compact. Then its underlying space is a finite collection of oriented points. However, as some curves are multiply covered, and so have isotropy groups, we must treat this as a non-effective 0 dimensional oriented orbifold. The contribution of each curve $[u]$ to the Gromov–Witten invariant $\int_{[\mathfrak{X}]^{vir}} 1$ is $\frac{\epsilon([u])}{|\Gamma([u])|}$, where $\epsilon([u]) = \pm 1$ is the orientation of the point $[u]$ and $|\Gamma([u])|$ is the order of the isotropy group $\Gamma([u])$ of $[u]$; in the McDuff–Wehrheim setup this is explained in [23, Section 5]. In the setup of Fukaya–Ono [12] we may readily calculate to get the same thing, taking the Kuranishi neighborhoods $(V \times W, E \oplus W, \Gamma)$ above at the isolated points of $\mathfrak{X}$; in the setup of Pardon it follows from the description of the virtual class by a single chart, [32, Section 2.3.3], as $1/\#\Gamma$ times the signed count of the Γ-set of solutions.

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